

“Propose and Review”: Interactive Bias Mitigation of Machine Classifiers

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Abstract

We develop a solution framework for mitigating algorithmic bias in machine-learning **classifiers**. We consider an interactive problem setting where Alice (e.g., the enterprise) proposes to use data records to build a target classifier, while Bob (e.g., the regulator) builds a **bias-proof classifier** to protect sensitive dimensions from being discriminated on. Specific measures of algorithm fairness and a priori definition of bias terms are exempted in this *propose and review* scenario. Our solution relies on the idea that the discriminating strength (i.e., the bias) in Alice’s target classifier corresponds to the separating capability (i.e., the signal) of Bob’s discrimination-proof classifier, thus the goal is to reduce the separability of Bob’s classifier while preserving the separability of Alice’s classifier, obviating the *fairness-utility tradeoff* by separating two classifications. We measure the pairwise distance between data attributes from indicating their differences in contributing to classifier’s separability. Multidimensional scaling (MDS) on the distance matrix helps establish the mapping between attributes and inferred bias sources. This bias-concentration information helps establish a ranking on attributes that suggests their bias-proneness. We mitigate bias at Bob’s classifier (i.e., limit the classifier’s separability) by searching for appropriate transformation of data records at bias-prone attributes. We consider polynomial/re-binning transformation at numerical/categorical data attributes, greedily suppressing attributes’ maximum bias concentration below a threshold. The framework is demonstrated with a detailed case study at a well-known classification dataset. We achieve good bias-mitigation performance at 10/12 machine classifiers; results are robust across multiple design choices and engineering options. We further showcase the framework’s performance at a larger dataset having > 100 attributes. Results document a clear example where classification’s fairness and classifier’s utility under certain definitions are both enhanced.

Keywords— algorithmic bias, fairness, machine-learning classifier, bias mitigation, data transformation

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1 Introduction

While discriminatory issues permeate human decision-making (e.g., *Bertrand and Mullainathan*, 2004), algorithm-based decision-support systems also demonstrate substantial bias (e.g., *Xiao and Benbasat*, 2015; *Ahsen et al.*, 2019; *Fu et al.*, 2021), despite the hope that algorithms can be used to attenuate bias in decision-making (*Ganju et al.*, 2020; *Zhang et al.*, 2021). Algorithmic biases are observed in facial recognition systems (*Raji and Buolamwini*, 2019), recommendation systems (*Schnabel et al.*, 2016), advertisement seeding systems (*Mehrabian et al.*, 2021) etc, bringing substantial negative effects to daily lives (*Crawford*, 2013; *Barocas and Selbst*, 2016; *Osoba and Welser*, 2017; *Howard and Borenstein*, 2018; *Lee et al.*, 2019).

Problem is pronounced with risk assessment tools that facilitate the solution of *prediction policy problems* (*Kleinberg et al.*, 2015). The fact that machine-learning models assessing individual or organizational agent’s risks, e.g., risk in healthcare outcomes (e.g. *Obermeyer and Lee*, 2017; *Obermeyer et al.*, 2019), risk of committing criminal behaviors (e.g. *Lum*, 2017; *Berk et al.*, 2021), or risk in economic performances (e.g., *Twala*, 2010), may be subject to considerable algorithmic biases (*Cirillo et al.*, 2020), becomes a significant issue in computational social sciences (*Lazer et al.*, 2020).

The phenomenon has drawn growing attention due to the rapid adoption of machine learning in digital decision-making (*Brynjolfsson and McElheran*, 2016; *Benjamin*, 2019). We identify considerable human-induced bias during our interaction with the machine, e.g., recommendation bias from sponsorship (*Xiao and Benbasat*, 2015) or bias due to cognitive imperfectness in interpreting machine-generated outcomes (*Ahsen et al.*, 2019). A larger concern lies around the biases that are less human-induced but data-induced: machines trained with data that encode human bias will reproduce, instead of eliminate, the bias; machines will only be as objective as the data used to train them (*Lum*, 2017). On digital platforms, such discriminatory bias is prevailing, in the peer to peer (P2P) setting as well as in the business-to-customer (B2C) setting (*Gunarathne et al.*, 2021). Mitigating algorithmic bias has become an important issue for machine-aided decision-making, although counter-intuitively, in certain cases mitigating algorithmic bias may lead to worse outcomes (*Zhang et al.*, 2021).

These concerns trigger the discussion around the *fairness* of machine-learning outcomes. This notion of *fairness* consists of two perspectives: (1) the fairness of tools in providing utilities (e.g., speech recognition systems may not work equally well for different demographic groups (*Koenecke et al.*, 2020)), and (2) the fairness of algorithms in terms of differentiating agents (e.g., recidivism prediction tools may assign higher risk rates to certain racial groups (*Lum*, 2017)). While the former issue can be eased considerably with better data-analysis practice, e.g., preventing selection bias in the training dataset via fairness-aware data collection (*Holstein et al.*, 2019), the latter issue is difficult to address.

The problem gathered considerable research efforts during the past decade. In general, people explore three approaches to reduce algorithmic bias and enhance fairness in machine-learning outcomes (*Hajian et al.*, 2016; *Fu et al.*, 2021; *Berk et al.*, 2021): *pre-processing*, *in-processing*, and *post-processing* (Table 1). Despite their effectiveness in different settings, each approach may demonstrate certain drawbacks:

- At *pre-processing*, people engineer input data records strategically, such as eliminating problematic records, changing the labels of certain records, imposing weights on records, adding noise, or expanding the dataset with more entries to counterweight (*Calders et al.*, 2009; *Kamiran et al.*, 2010; *DeBrusk*, 2018).

Drawbacks: In practice, this approach often requires case-by-case examination of datasets, which are of varied quality. The effectiveness of specific tactics can be difficult to generalize to different machine-learning models.

- At *in-processing*, people develop fairness-aware algorithms (*Kamiran et al.*, 2010; *Bechavod and Ligett*, 2017; *Friedler et al.*, 2019; *Ahsen et al.*, 2019) that incorporate the consideration of fairness in algorithm design (e.g, adding fairness terms in the objective function to be optimized), based on specific fairness metrics.

Drawbacks: In-processing is not straightforward for existant algorithms and may require re-design of

models or workflows. The design depends on specific metrics of fairness (see below).

- At *post-processing*, people re-engineer algorithm outcomes after results are carried out, based on regression analysis (Ayres, 2010; Lakkaraju et al., 2017; Jung et al., 2018) or various heuristics (Vyas et al., 2020).

Drawbacks: Engineering machine outcomes leaves the model untouched, and biases remain in the classifier. It re-opens the space for substantial human intervention and counters the utilization of algorithmic intelligence in the first place. Regression analysis may depend heavily on specification (Fu et al., 2021).

Approach	Operation	Drawback
<i>Pre-processing</i>	fair-aware data collection data record manipulation	case-by-case examination of dataset not applicable to unknown bias
<i>In-processing</i>	fair-aware algorithm design	not straightforward for existing algorithms dependent on metrics of fairness
<i>Post-processing</i>	output filtering, bias regression	bias unremoved from model not applicable to unknown bias
Proposed solution		
Approach	Operation	Highlight
<i>Pre-possessing</i>	attribute-bias mapping & estimation bias concentration determination bias mitigation via data transformation	applicable to unknown bias applicable to existing algorithms transfer of mitigation strategies across models

Table 1: Approaches for algorithmic bias mitigation and fairness enhancement of machine-learning outputs.

A fundamental issue complicates this problem: the notion of *fairness* of machine-learning outcomes is not well-defined. There are many definitions of algorithm fairness (e.g., Hardt et al., 2016; Barocas et al., 2017; Chouldechova and Roth, 2018), and correspondingly many assessment tools (e.g., Aequitas (Saleiro et al., 2018), AI Fairness 360 (Bellamy et al., 2018)) developed to evaluate system’s concentration of fairness. Certain fairness metrics correlate with each other, while considering different statistical properties of model output (Friedler et al., 2019). However, well-adopted fairness measures can be contradictory, and it is often difficult and even mathematically unlikely to simultaneously achieve fairness under multiple definitions (Chouldechova, 2017; Courtland, 2018; Berk et al., 2021). For example, it is impossible to achieve three popular notions of algorithm fairness - *predictive parity*, *equal false-positive rates*, and *equal false-negative rates* - at the same time (Kleinberg et al., 2017). As a result, bias-mitigation solutions based on certain fairness metrics may be invalid under other metrics. Even more troublesome, it is shown that using the proposed fair algorithm upon a specific fairness definition can actually make everyone worse off, including the exact class of people the algorithm aims to protect (Fu et al., 2021). These observations suggest that deriving bias-mitigation solutions based on fairness metrics, at either *pre-processing*, *in-processing*, or *post-processing*, may be subject to fundamental system error.

Besides lacking consensus on the fairness measure, another latent issue complicates the problem. So far, most extant solutions work on pre-existing bias attributes, i.e., gender, age, or race; the bias can be identified in various ways such as adopting the 80% rule, benchmarking, outcome test, or regression analysis (Fu et al., 2021). Yet people acknowledge that other attributes may be correlated with these biased attributes (Calders et al., 2009), so removing these attributes from model input is insufficient (Pedreshi et al., 2008). For example, after the removal of “race” from input entries, a trained predictor may still assign a higher crime rate to black people (Mehrabani et al., 2021), suggesting that other attributes may still embody racial bias in a latent manner. Such could be the case at attribute “annual income”: as black people on average earn less than other racial groups, the inclusion of “annual income” may introduce racial bias to the classifier even when “race” is excluded. Indeed, the phenomenon that seemingly neutral attributes can be a serious source of algorithmic bias due to their proxy role for sensitive attributes, has been discovered in different contexts (Obermeyer et al., 2019). Aware of this, researchers have designed intelligent methods to re-engineer input attributes to eliminate their association with biased attributes (Feldman et al., 2015; Johndrow and

Lum, 2019; Fu et al., 2021).

Nevertheless, the assumption of bias attributes’ pre-existence needs to be relaxed. Although so far in most cases people agree on sensitive attributes such as gender and race, it is rather difficult to identify potential source of bias at other attributes. Is your geological location, middle school choice, or even breakfast expenditure, being discriminated upon? Few consider them currently, but they could be just as discriminatory as race and gender. More so, a source of bias may concern multiple attributes, instead of leaning on a single attribute. For example, “people with critical credit history, residence since 2.8 years at least, with savings at most for 100 units, and with no checkings, will be assigned a lower credit score” (Pedreshi et al., 2008). In this case, we know that a certain bias term exists, but we don’t know it concerns which attribute(s). Even more complicated is when we don’t even know if there is indeed any bias in a constructed machine classifier, nor what is the nature of each bias term, but we want that no bias exists in that attribute space: “these personal information clauses are used to predict individual’s tendency to vote in an election; we don’t know if there will be bias in the predictor, but we don’t want people to be biased upon along any dimension.”

The above discussion suggests an alternative research angle: since (1) fairness of machine-learning outcomes is ill-defined and (2) the identification of discriminatory terms can be subjective and may evolve over time, the problem of achieving algorithm fairness in decision-support systems is better rephrased as a design question (Hevner et al., 2004; Gregor and Hevner, 2013; Rai, 2017): we do not attempt to define how predictions are fair and which attributes contain bias; instead, we focus on developing a general framework for the strategic mitigation of bias in machine-learning classifiers along any suggested discriminatory dimension.

Bias mitigation is to be cast in a practical scenario when considering that bias terms are not a priori known. People are often vague about specific source of discrimination, and it is desired that a machine classifier is simply devoid of any possible sensitive concern along a certain dimension (“just let the machine be *fair* on this matter”), even if the model is not classifying in that dimension. For example, the government may desire that a company’s commercial predictor on individual’s credit card default rate should not bias towards people having different education years. This is exactly because a person’s education background may be correlated with his credit card default rate (the correlation could be either high or low, though); but for social welfare considerations, this correlation needs to be overlooked in credit card approvals.

Intrinsically, this problem can be considered in an “attack-defense” scenario, with respect to two stakeholders: Alice (i.e., the enterprise) is asking to use data records having certain attributes (e.g., personal information clauses) to develop machine classifiers for commercial usage, and Bob (i.e., the regulator) is responsible to ensure that, based on Alice’s usage of data, her classifier should not demonstrate substantial discriminatory strength along a certain sensitive dimension. The “attack-defense” scenario then translates into a “*propose and review*” scenario. As social norms evolve over time, enterprises and regulators may constantly go through such *propose and review* procedures to ensure legal use of personal information in machine algorithms. Realistically, under this mechanism, *private information is held on both sides*: Bob does not have access to Alice’s models and algorithms (since regulators should fully respect firms’ intellectual properties), while Alice does not have accurate knowledge on the sensitive dimension that Bob concerns based on her proposed usage of data (otherwise she could tailor the algorithms to achieve fairness on that dimension).

In this study, we develop a framework to functionalize this *propose and review* mechanism for algorithmic bias mitigation (Figure 1). Our method approaches the problem via the transformation of data records, and enables the transfer of bias-mitigation strategies across machine classifiers (i.e., from Bob to Alice). We do not assume a priori knowledge of bias attributes while infer sources of discrimination directly from data, relying on a core idea that the discriminatory strength (i.e., the bias) in Alice’s target classifier corresponds to the separating capability (i.e., the signal) of Bob’s discrimination-proof classifier. Attributes’ contribution to classifier’s separating capability is measured by the difference in the separability of data when attributes are included in versus excluded from model input. A distance measure between pairs of attributes is constructed based such differences computed at different attribute inclusion. Applying multidimensional scaling (MDS) on the distance matrix, we establish the mapping between attributes and inferred bias sources. This bias-concentration information helps establish a ranking on attributes that suggests their bias-proneness. We adopt a greedy method that iteratively transforms data records at the most bias-prone attribute. Polynomial transformation applies to numerical attributes; re-binning transformation applies to

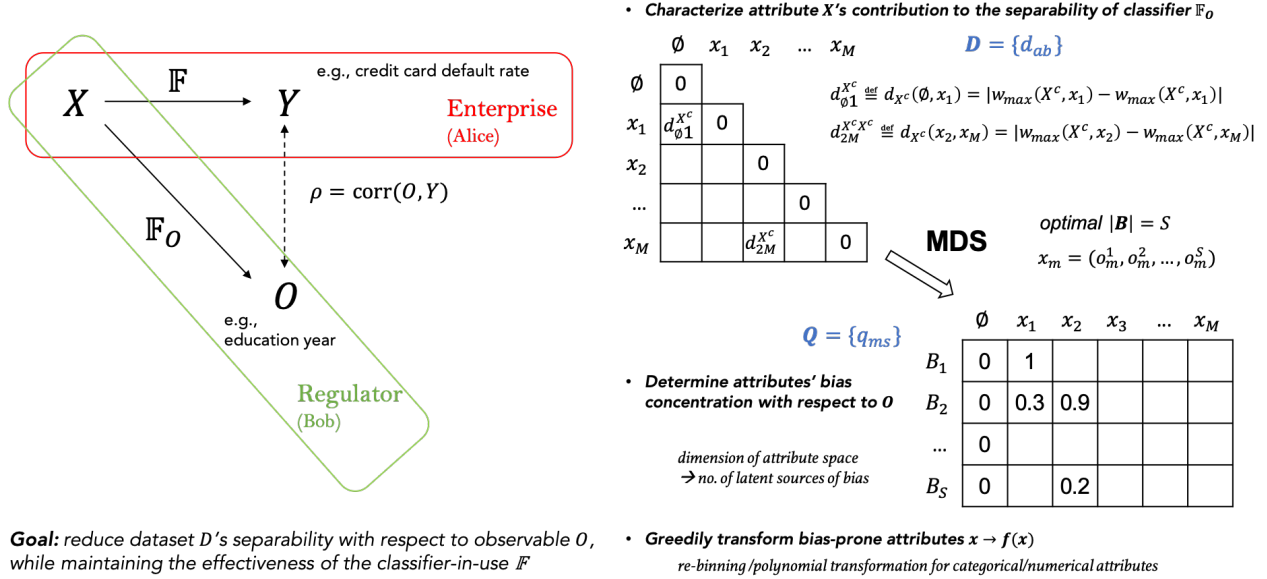


Figure 1: Illustration of the *propose and review* solution framework for algorithmic bias mitigation.

categorical attributes. Transformation of bias-prone attributes makes sure that bias concentration at the discrimination-proof classifier is subdued below an acceptance threshold.

We validate the framework with a well-known classification dataset (UCI adult dataset (Dua and Graff, 2019)). The detailed case study suggests that our solution achieves good bias-mitigation performance across a panel of machine-learning classifiers. The solution obviates the *fairness-utility tradeoff* (Feldman et al., 2015): it substantially reduces bias concentration of the observing classifier, i.e., at Bob’s protected dimension, while largely preserves the separability of the main classifier, i.e., at Alice’s usage dimension. The results are robust against a panel of design choices and engineering options. We further showcase the framework’s performance at a larger dataset (HCDR dataset (HCDR dataset, 2018)). Results demonstrate a complete override of the tradeoff, where fairness and utility under certain definitions are both improved.

Organization of the paper

The rest of this paper is organized as follows. In Section 2, we review recent literature on algorithmic bias that bring insights to this study. In Section 3, we establish the theoretical setup for our bias-mitigation framework, outline the method’s workflow. We demonstrate the method’s performance with a detailed case study in Section 4, studying a well-known classification dataset. In Section 5, we conduct a panel of robustness checks for the main result, considering different design options. The method’s performance at a larger dataset having > 100 attributes is shown in Section 6. Limitations, future directions, and managerial implications of this study are discussed in Section 7, where the paper is concluded.

2 Related Literature

Algorithmic bias emerges as an important topic in management research during the past few years. Existant studies hinge on pre-determined bias attributes such as gender and race, i.e., the “protected attributes”. A number of studies deliver important insights that inform our research ideas.

Calders et al. (2009) and Kamiran et al. (2010) investigated a number of pro-processing methods for bias mitigation, e.g., changing the labels of data records (“massaging” (Calders et al., 2009), “leaf relabeling” for decision trees (Kamiran et al., 2010)) or imposing weights on records (“reweighing” (Calders et al., 2009)). They considered the situation where there are correlations between attributes and the class label, instead

of between attributes themselves. This suggests hierarchical (greedy) mitigation, where the most sensitive attribute is used as the class label, and we determine correlation between it and other attributes; the analysis is then conducted on the second most sensitive attribute, and so on. This requires a ranking of attributes’ *sensitiveness*, which may solicit human judgment.

Feldman et al. (2015) defines “disparate impact” according to the 80% rule advocated by the US Equal Employment Opportunity Commission ¹. Given dataset D , protected attributes X , remaining attributes Y , and binary class label C , the *disparate impact removal* problem is to take D and return an engineered dataset $\bar{D} = (X, \bar{Y}, C)$ that can be certified as not having disparate impact, by only changing $Y \rightarrow \bar{Y}$. The problem is formulated in the bi-lateral scenario where Alice runs an algorithm to remove disparate impact from D , and Bob is responsible to verify that D indeed has no disparate impact. A verification method and a bias-removal method are designed coordinately. The verification method regards X as the prediction class and uses $\bar{Y}$ to predict X . If X is poorly predicted by $\bar{Y}$, then it is declared that the cleaned $\bar{D}$ is free from disparate impact. Acknowledged by *Feldman et al.* (2015), there is always a *fairness-utility tradeoff*: increasing the fairness of the classifier will degrade its classification capability.

Ahsen et al. (2019) proposed a bias-aware classification algorithm for breast cancer diagnosis. The model adopts the Gaussian assumption of bias terms, considering both machine-induced bias and cognition-related bias. Optimal values of algorithm parameters are derived in closed forms. The analysis may not generalize to cases having alternative bias structures besides Gaussian.

Lambrecht and Tucker (2019) reported a field test on algorithm’s performance in delivering job-promoting ads that are intended to be gender-neutral. They found that fewer women saw the ad than men since younger women are a prized demographic and are more expensive to show ads to, so that an algorithm that simply optimizes cost-effectiveness in ad delivery will deliver ads that were intended to be gender-neutral in an apparently discriminatory way. This empirical regularity extends to a few major digital platforms.

Samorani et al. (2021) studied algorithm bias in medical appointment scheduling on different racial groups. They demonstrated that state-of-the-art scheduling systems can cause black patients to wait about 30% longer than non-black patients, a bias not eliminated by removing socio-economic indicators from the data. They proposed new objective functions that minimize both schedule cost and racial disparity. The race-aware objective minimizes the waiting time of the racial group expected to wait the longest.

Building on *Feldman et al.* (2015) and *Johndrow and Lum* (2019), *Fu et al.* (2021) extends an intelligent bias-removal method that replaces absolute values of data records Y by their quantiles $\bar{Y}$ within each category of the protected attribute. Specific treatments are derived for attributes of various types: continuous, categorical, ordinal and mixed. The bias-mitigation method is tested on a number of experiments and yields good performance.

Kallus et al. (2022) considered a problem setting similar to our study, where sensitive attributes A are protected and a set of observables Z are used as the proxies. They assume a main dataset (Z, Y) where Y is the outcome, and an auxiliary dataset (A, Z) embedding the relationship between A and Z . They developed a tool that can compute the tightest possible bounds on disparity that could possibly be learned from the combined data, i.e., the fundamental limits of identification, along a panels of disparity measures.

3 Theoretical Setup

We consider the *propose and review* scenario between Alice (e.g., the enterprise) and Bob (e.g., the regulator). Alice proposes to use dataset D having N records and M attributes $X = \{x_1, x_2, \dots, x_M\}$ in each record. D will be input into Alice’s machine-learning model $\mathbb{F}$ that classifies on variable Y which has C class, i.e., $Y = \{y_1, y_2, \dots, y_C\}$: ²

$$\mathbb{F}(D, X) = \mathbb{F}(x_{1,i}, x_{2,i}, \dots, x_{M,i}) \rightarrow \tilde{y}_i, \quad \text{for } i = 1, 2, \dots, N. \quad (1)$$

¹Equal Employment Opportunity Commission (1979), <https://www.eeoc.gov/laws/guidance/questions-and-answers-clarify-and-provide-common-interpretation-uniform-guidelines>

²In this study, we focus on binary Y , then demonstrate a method that extends the analysis to multi-nary Y .

Bob reviews Alice’s proposal (D, X, Y) . Bob aims to reduce the discriminating capacity of (D, X) at a specific dimension, determined by an observable O not identical to Y .³ For example, for social welfare purposes, Bob may request that Alice’s commercial algorithm of a housing insurance scoring system should not bias against people having different education backgrounds. The length of school years, nevertheless, is believed to be correlated with people’s payment capability, but this is exactly a relationship Bob desires to overlook. This correlation between Alice’s target variable Y and Bob’s selected observable O , denoted by $\rho = \text{corr}(O, Y)$, can be utilized in bias mitigation.⁴

Bob examines the discriminating capacity of (D, X) not directly on Alice’s classifier $\mathbb{F}$ which is not transparent to Bob, but on an observation classifier $\mathbb{F}_O$ he constructs:

$$\mathbb{F}_O(D, X) = \mathbb{F}_O(x_{1,i}, x_{2,i}, \dots, x_{M,i}) \rightarrow \tilde{o}_i, \quad \text{for } i = 1, 2, \dots, N. \quad (2)$$

Considering the protected dimension O , sources of bias in the data D that lead to the discrimination at O are identified when analyzing attributes X ’s contribution to $\mathbb{F}_O$ ’s separation capacity. Based on this identification, encoded as an attribute-bias concentration matrix, bias mitigation proposes to engineer surfaced attributes, leading to $X \rightarrow X'$. Each treated attribute undergoes an optimized transformation $x_m \rightarrow x'_m$.⁵

These transformations on input attributes are approved via Bob’s $\mathbb{F}_O$, and are transferred to Alice’s $\mathbb{F}$. The transfer is valid because the method only pre-processes model inputs, which are the same for $\mathbb{F}$ and for $\mathbb{F}_O$, without interfering in the classifiers’ algorithms. This approach inherently embodies the idea that the causal relationships between variables (i.e., input attributes) are key to address algorithm fairness (*Kusner and Loftus, 2020*), irrespective of specific algorithmic operations.

Specifically, our solution framework consists of the following theoretical components.

3.1 Bias-signal duality

Bob requests that Alice’s usage of attributes X should not discriminate with respect to observable O , along any dimension in $\mathbb{F}_O$ ’s model space. This leads to a core idea: the unwanted bias in $\mathbb{F}$ corresponds to the signal in $\mathbb{F}_O$. The condition that X does not demonstrate bias with respect to O , is equivalent to the situation where dataset D is not *separable* under $\mathbb{F}_O$. Therefore, the goal of bias mitigation translates to the goal of partially limiting the *separating capability* of attributes: we develop strategic engineering on input attributes X , $X \rightarrow X'$, such that the transformed dataset D' has reduced separability in the $\mathbb{F}_O$ space, while maintaining sufficient separating capability in the target $\mathbb{F}$ space.

This points to the **fairness-utility tradeoff**: increasing the fairness of the classifier will degrade its classification capability (*Feldman et al., 2015*). We partially obviate this tradeoff by considering two related classifiers, one for usage and the other for observation. Fairness of the observation classifier is achieved, via degrading its classification capability as much as possible. The target classifier’s classification capability is largely preserved, although we expect it to also go down, but not as significantly. We aim to develop a method that achieves the dual goal of subduing bias while preserving signal.

Demonstration of the method’s effectiveness asks for a fairness metric and a classification performance metric, while our framework itself does not anchor on any fairness metric or classification performance metric. We characterize the generic bias-signal duality entirely from data.

3.2 Latent bias source

Suppose a set of S bias sources embedded in the data D , $\mathbf{B} = \{B_1, B_2, \dots, B_S\}$, which lead to discrimination at O . A certain bias source B_s may be associated with one or multiple attributes in X , since attributes are often not independent variables. To characterize these dependencies, we establish a mapping between

³We term O as an *observable* instead of an “instrument variable”; it is not necessary that O and Y are highly correlated.

⁴ ρ can be estimated from external analysis; it is not necessarily a large value.

⁵Different transformations can apply, such as adding noise to x ’s records, removal of x from X , linear/polynomial transformation of x ’s records. In this study, we consider re-binning of data records when x is a categorical variable, and polynomial transformation of records when x is a numerical variable.

attributes and bias sources, represented with matrix $\mathbf{Q} = \{q_{ms}\}^{M \times S}$, where q_{ms} indicates attribute x_m 's concentration of bias B_s . Bias concentration matrix $\mathbf{Q}$ facilitates a straightforward exposition of data attributes with respect to origins of discrimination, based on which we determine strategic bias-mitigation policies.

Another core idea of our solution is that we relax the assumption that biases $\mathbf{B}$ are pre-determined and are externalized on attributes. In many cases, sensitive attributes are known beforehand (e.g., gender or race); yet the source of discrimination may associate with multiple attributes, and in the long term may evolve with social norms, asking us to consider more general scenarios.

According to the nature of a bias term, four scenarios may occur:

- ◊ (i): A bias term B_s is known, and is an attribute x_m in X .
 → e.g., "Certain racial groups have more tendency to commit crimes." Race is an explicit data attribute. In this most straightforward case, $q = 1$ on x_{race} .
- ◊ (ii): A bias term B_s is known, but not a single attribute in X .
 → e.g., "People with critical credit history, residence since 2.8 years at least, with savings at most for 100 units, and with no checkings, will be assigned a lower credit score (*Pedreshi et al.*, 2008)." This bias concerns multiple data attributes. In this case, $q \neq 0$ for these attributes.⁶
- ◊ (iii): For a set of data attributes X and a classifying dimension O , a bias term B_s that discriminates at O is unknown, but we know it exists.
 → e.g., "Some people have more tendency to drop school." In this case, $q \neq 0$ for some attributes (to be identified).
- ◊ (iv): For a set of data attributes X and a classifying dimension O , we don't know whether biases exist or not, and if they do, how many independent discriminatory sources are there.
 → e.g., "These personal information clauses are used to predict individual's likelihood to vote. We don't know if there will be bias in the machine classifier, nor what is the nature of each discriminatory term." But we want to guarantee that everyone has the equal right to vote, i.e., the dataset is completely void of bias with respect to voting likelihood.

In this study, we deal with Scenario (iv), the most general case. In this scenario, we don't consider bias terms in the absolute sense, i.e., gender or race is a *absolute* bias attribute, but in the relative sense, i.e., consider the source of discrimination as biasing at a specific classifying dimension O .

3.3 Attributes' separating capacity

We indicate the existence of potential bias source in the attribute space by measuring attributes' separating capacity. We desire to reduce the dataset's separating capability at $\mathbb{F}_0$; in this case, $\mathbb{F}_0$'s classification power encodes the source of discrimination. We consider the bias concentration of $\mathbb{F}_0$ as its maximum separating capability w_{max} . The larger w_{max} is, the greater $\mathbb{F}_0$ is separating (i.e., discriminating). A set of attributes X^* 's contribution to $\mathbb{F}_0$'s maximum separating capability is denoted as $w_{max}(X^*)$.

We define w_{max} in the following way. Recall the original dataset D having N records and M attributes; consider each data record i associated with a binary class label $o^i \in \{+, -\}$. Data records on an attribute $x_m = \{x_{m,1}, x_{m,2}, \dots, x_{m,N}\}$ is divided into the two classes, $\{x_{m,i}^+\}$ and $\{x_{m,i}^-\}$. We calculate the mean value of attributes, and normalize each attribute in the min-max manner (*Jain and Bhandare*, 2011), i.e., $x_{m,i}^{\text{norm}} = (x_{m,i} - x_m^{\min}) / (x_m^{\max} - x_m^{\min})$. After the transformation of data records, different attributes are comparable in the data space.

The maximum separating capacity w_{max} of a set of attributes X^* collects the contribution from each attribute

⁶ q may still be non-zero on other attributes.

$x_m \in X^*$, denoted as S_m :

$$w_{max}(\{X^*\}) = \|S_m\|_\alpha = \sqrt{\frac{S_1^2 + S_2^2 + \dots + S_{|X^*|}^2}{|X^*|}}, \quad \text{for } m = 1, 2, \dots, |X^*|, \quad (3)$$

where the α is the vector norm, $|X^*|$ is the number of elements in X^* . We consider $\alpha = 2$, the L2 norm.

The separating capacity S_m of individual attributes is defined respectively for numerical attributes and for categorical attributes:

- When x_m is a numerical attribute, S_m is defined by the difference between the centroid of positive class labels and negative class labels, i.e., $S_m = |\bar{x}_m^+ - \bar{x}_m^-|$, where $\bar{x}_m^{+/-}$ is the average value of the positive and the negative class, respectively. The idea is to indicate the separation by the difference in positive and negative class labels' mean.
- When x_m is a categorical attribute, S_m is defined by the average frequency difference between the positive and the negative class of each category, i.e., $S_m = \frac{1}{K} \sum_{k=1}^K \left| \frac{N_k^+}{N^+} - \frac{N_k^-}{N^-} \right|$, where K is the number of categories under attribute x_m , $N_k^{+/-}$ is the frequency for the k -th category with positive/negative class, and $N^{+/-}$ is the sum of the frequency for the positive/negative class. The idea is to characterize the difference in positive and negative class labels' distribution.

Overall,

$$S_m = \begin{cases} |\bar{x}_m^+ - \bar{x}_m^-|, & \text{numerical} \\ \frac{1}{K} \sum_{k=1}^K \left| \frac{N_k^+}{N^+} - \frac{N_k^-}{N^-} \right|, & \text{categorical} \end{cases}. \quad (4)$$

The above definition of maximum separating capacity w_{max} considers binary classification. The definition extends to multi-nary classification, with S_m defined as the maximum value among all possible two-class pairs.

When the classified dimension O is multi-nary with class labels $\{C_1, C_2, \dots, C_V\}$, the maximum separating capacity w_{max} is defined as

$$w_{max}(\{X^*\}) = \max_{\forall C_p, C_q, p \neq q} \left\{ \left\| S_i^{(C_p, C_q)} \right\|_\alpha \right\}, \quad \text{for } p, q = 1, 2, \dots, V. \quad (5)$$

3.4 Attribute-bias concentration

It is rare that a dataset is completely inseparable under $\mathbb{F}_0$, thus we consider that there is always potential source of discrimination in the attribute space. We deduce the number of bias sources directly from data, considering the most general case of bias embedding (Scenario (iv)). We adopt multidimensional scaling (MDS) (Cox and Cox, 2008; Mair et al., 2016; Saeed et al., 2016) to derive the optimal number of bias sources $S = |\mathbf{B}|$. Given a distance matrix $\mathbf{D} = \{d(x_i, x_j)\}$ for a set of objects $x_1, x_2, \dots, x_W$, MDS minimizes the cost function, termed as *stress*, to obtain an appropriate set of coordinates for the objects, which best match objects' distance measures $d(x_i, x_j)$:

$$\text{stress}_{\text{MDS}}(x_1, x_2, \dots, x_W) = \sqrt{\sum_{i \neq j=1, \dots, W} (d(x_i, x_j) - \|x_i - x_j\|)^2}. \quad (6)$$

To conduct MDS, we establish the distance measure $d(x_a, x_b)$ between two attributes x_a and x_b based on the maximum separating capability w_{max} . This metric facilitates the determination of attributes' distance by considering the difference in attributes' concentration of bias while classifying on O , i.e., difference in attributes' contribution to $\mathbb{F}_0$'s separating capability.

For an attribute x_m , the separating capability of $\mathbb{F}_0$ is different when x_m is excluded from model input, from when it is included. The contribution of x_m to the classifier's separating capability – i.e., x_m 's *bias concentration* – can be indicated by such difference, $|w_{max}(X) - w_{max}(X/x_m)|$.

The pair-wise distance $d(x_a, x_b)$ between attributes x_a and x_b is then computed as the difference in w_{max} across the two scenarios where either x_a or x_b is excluded. For other attributes besides x_a and x_b (i.e., the companion attributes), they are either excluded or not; this gives the definition of sub-distance measures $d_{X^c}(x_a, x_b)$:⁷

$$d_{X^c}(x_a, x_b) \triangleq |w_{max}(\{X^c, x_a\}) - w_{max}(\{X^c, x_b\})|. \quad (7)$$

For example, when only x_1 is additionally included in model input, the distance measure between x_a and x_b is $d_{x_1}(x_a, x_b) = |w_{max}(\{x_1, x_a\}) - w_{max}(\{x_1, x_b\})|$. Similarly, when only x_a or x_b are excluded and all other attributes are included, the distance is $d_{X/\{x_a, x_b\}}(x_a, x_b) = |w_{max}(\{X/x_b\}) - w_{max}(\{X/x_a\})|$.

The complete enumeration of these sub-distance measures d_{X^c} for $\forall X^c \in X/\{x_a, x_b\}$ echos the computation of the SHAP value (Lundberg and Lee, 2017; Lundberg et al., 2020), which may require factorial complexity. We regard this enumeration at different levels: at level 0, $X^c = X/\{x_a, x_b\}$ and $|X^c| = |X| - 2$; at level 1, X^c is $X/\{x_a, x_b\}$ excludes any extra attribute besides x_a and x_b , thus $|X^c| = |X| - 3$ and there are $C_1^{|X|-2} = |X| - 2$ options of X^c ; at level h , X^c is $X/\{x_a, x_b\}$ excludes any h extra attribute besides x_a and x_b , thus $|X^c| = |X| - 2 - h$ and there are $C_h^{|X|-2}$ options of X^c .

The overall distance measure summarizes sub-distance measures up to level H :

$$\begin{aligned} d_D^H(x_a, x_b) &\triangleq \frac{d_{X/\{x_a, x_b\}} + \sum_{x_m^1 \in X/\{x_a, x_b\}} d_{X/\{x_a, x_b, x_m^1\}} + \dots + \sum_{\{x_{m_1}^H, \dots, x_{m_H}^H\} \in X/\{x_a, x_b\}} d_{X/\{x_a, x_b, x_{m_1}^H, \dots, x_{m_H}^H\}}}{C_0^{|X|-2} + C_1^{|X|-2} + \dots + C_H^{|X|-2}} \\ &= \frac{\sum_{h=0}^H \sum_{\{x_{m_1}^h, \dots, x_{m_h}^h\} \in X/\{x_a, x_b\}} d_{X/\{x_a, x_b, x_{m_1}^h, \dots, x_{m_h}^h\}}}{\sum_{h=0}^H C_h^{|X|-2}}, \end{aligned} \quad (8)$$

upon which the distance matrix $\mathbf{D} \triangleq \{d_{ab}\} = \{d_D(x_a, x_b)\}$ is constructed.

We consider an additional row and column in $\mathbf{D}$ for the distance between when an attribute x_m is excluded and when no attribute is excluded: $d_{X^c}(\emptyset, x_m) \triangleq |w_{max}(\{X^c, x_m\}) - w_{max}(\{X^c\})|$. The corresponding

$$d_D^H(\emptyset, x_m) \triangleq \frac{\sum_{h=0}^H \sum_{\{x_{m_1}^h, \dots, x_{m_h}^h\} \in X/\{x_m\}} d_{X/\{x_m, x_{m_1}^h, \dots, x_{m_h}^h\}}}{\sum_{h=0}^H C_h^{|X|-1}}. \quad (9)$$

This imposes an origin $\emptyset$ in the attribute space⁸, and attributes' (raw) distance to origin (i.e., their contribution to classifier's separating capability) can be compared. Inclusion of the origin $\emptyset$ in the distance matrix removes the translational symmetry of the MDS.

MDS is conducted on the distance matrix $\mathbf{D}$, and we use the elbow plot to derive the optimal cardinality S of bias sources encoded in the model space.⁹ Each attribute's loading o_m^s on the specific bias coordinate (computed via MDS: $\hat{x}_m = (o_m^1, o_m^2, o_m^3, \dots, o_m^S)$) is used to determine the entry q_{sm} in the bias concentration matrix $\mathbf{Q}$, whose row dimension is S .¹⁰

⁷To write out the equation: $d_{X^c}(x_a, x_b) = |(w_{max}(X^c, x_a, x_b) - w_{max}(\{X^c, x_b\})) - (w_{max}(X^c, x_a, x_b) - w_{max}(\{X^c, x_a\}))| = |w_{max}(\{X^c, x_a\}) - w_{max}(\{X^c, x_b\})|$.

⁸The separating capacity w_{max} of the origin is similarly defined as

$$w_{max}(\{X, \emptyset\}) = \|\mathbf{S}_m\|_\alpha = \sqrt{\frac{S_1^2 + S_2^2 + \dots + S_M^2}{M}}, \quad \text{for } m = 1, 2, \dots, M. \quad (10)$$

⁹When $H = 0$, i.e., we only consider the topmost distance measure $d_{X/\{x_a, x_b\}}(x_a, x_b) = |w_{max}(\{X/x_b\}) - w_{max}(\{X/x_a\})|$, MDS always has a plain optimality at $S = 1$, since the distance measures are defined in the first place as the absolute differences of a set of scalar values. With the origin implemented to fix the shift, a unique set of optimal 1-D coordinates can be obtained.

¹⁰When the bias source is unknown, q_{sm} is o_m^s . When the bias source is known, for example, x_a is the bias attribute, a rotation transformation $\mathbf{R}^{S \times S}$ is applied on matrix $\tilde{\mathbf{Q}}^{S \times M} = \mathbf{R}^{S \times S} \mathbf{Q}^{S \times M}$, so that the bias attribute is projected onto the axis $s = 0$, i.e., $\tilde{q}_{0,a} \neq 0$ and $\tilde{q}_{i>0,a} = 0$. Then the row $s = 0$ of $\tilde{\mathbf{Q}}^{S \times M}$ is normalized by dividing $\tilde{q}_{0,a}$ and taking the absolute value. A normalized bias concentration matrix $\tilde{\mathbf{Q}}_{\text{norm}}^{S \times M}$ is obtained, informing how much concentration of the x_a bias do other attributes contain.

3.5 Bias mitigation

Attributes' bias concentration (i.e., contribution to the separating capability of $\mathbb{F}_0$) are recorded in matrix $\mathbf{Q}$, whose elements q_{sm} indicate the coordinates of attribute x_m at axis B_s in the bias hyperspace. The larger q_{sm} is, the larger the bias concentration is. An attribute's overall bias concentration across all bias axis is indicated by its (Euclid) distance to origin, $d_\emptyset(\emptyset, x_m)$, calculated on $\mathbf{Q}$:¹¹

$$d_\emptyset(\emptyset, x_m) = \sum_{s=1}^S (q_{sm} - q_{s\emptyset})^2, \quad \text{for } m = 1, 2, \dots, M. \quad (11)$$

This provides a ranking on attributes, with the attribute having the largest distance to origin being the attribute that has the largest bias concentration with respect to O .

We wish to restrict the maximum separating capability of $\mathbb{F}_0$ below a threshold ϵ .¹² This corresponds to suppressing attributes' distance to origin within the ϵ -radius ball. After the engineering on data records such that attributes' largest distance to origin falls below ϵ , i.e., $\max_{x_m \in X} d_\emptyset(\emptyset, x_m) < \epsilon$, Bob can allow the engineered data to be used by Alice in her $\mathbb{F}$.

We achieve this goal in the greedy manner, utilizing the above ranking $d_\emptyset(\emptyset, x_m)$. At each iteration, we pick up the attribute that has the largest distance to origin, and look for the appropriate transformation of data records at this attribute that suppresses its distance to origin below ϵ . After applying the transformation to this attribute, the distance matrix $\mathbf{D}$ is recalculated, the MDS is re-conducted, and we look at attributes' new distance to origin via the updated $\mathbf{Q}$. The iteration stops when all attributes' distance to origin fall below ϵ .¹³

We consider two distinct families of transformation f for numerical attributes and for categorical attributes.

- For numerical attributes, we consider polynomial transformation of each data record z at that attribute. We consider a single polynomial term of z at power α , i.e., $f(z) = z^\alpha$, instead of multiple terms (e.g., $f(z) = z^\alpha + z^{\alpha-1} + \dots$). In particular, there is

Proposition: Data record is invariant under linear transformation, $z' = \alpha z + \beta$.

Proof. According to the min-max normalization of data records, there is

$$z'_{\text{norm}} = \frac{(\alpha z + \beta) - (\alpha z^{\min} + \beta)}{(\alpha z^{\max} + \beta) - (\alpha z^{\min} + \beta)} = \frac{z - z^{\min}}{z^{\max} - z^{\min}} = z_{\text{norm}}. \quad (12)$$

□

To preserve the sign of data records, we consider odd integer (e.g., 3, 5, 7) and odd fraction (e.g., 1/3, 1/5, 1/7) powers. During the search of the appropriate transformation, the power increases until the condition $\max_{x_m \in X} d_\emptyset(\emptyset, x_m) < \epsilon$ is met.¹⁴

- For categorical attributes, we consider the re-binning of categories, and re-label each data record z at that attribute. For example, for an attribute having 5 categories, we may find that a specific 3-category re-binning of data records can satisfy the ϵ -radius condition for attributes' distance to origin. The search for appropriate re-binning follows a specific order, where we distinguish ordinal categories from non-ordinal categories (Appendix B). In particular, there is

¹¹A (raw) distance to origin is already captured as the element $d_D(\emptyset, x_m)$ in the distance matrix $\mathbf{D}$. MDS implements the optimal coordinates on $\mathbf{D}$; attributes' distance to origin are recalculated with the coordinates $q_{sm} \in \mathbf{Q}$. The updated distance is sharpened and is consistent with the structure of the attribute space, while not deviating too much from the raw distance.

¹² ϵ can be determined via the correlation ρ between variable O of Bob's observation classifier $\mathbb{F}_0$ and variable Y of Alice's target classifier $\mathbb{F}$. For example, $\epsilon = 1 - \rho(O, Y)$: the more O is correlated with Y , the less the bias concentration on $\mathbb{F}_0$ can be tolerated.

¹³It is possible that we apply transformation to the same attribute multiple times, as the transformation of a certain attribute may also influence the distance to origin of an already transformed attribute. In an extreme case, the transformation can iterate between two attributes, gradually bringing down their distance to origin. This iteration is not guaranteed to stop, though, and it is possible that no valid set of transformation on attributes can be found, so as to achieve the goal specified by ϵ .

¹⁴For example, we test z^3, z^5, z^7 etc. if the maximum distance to origin $d_\emptyset$ goes down as the power increases; if $d_\emptyset$ goes up as power increases, we test $z^{1/3}, z^{1/5}, z^{1/7}$ etc. For negative values of z , we transform the absolute value and preserve the sign.

Proposition: For binary attributes, removing the attribute is the single possible transformation.

Proof. For binary attributes (which is categorical), the only re-binning option is to merge the two categories into one single category. This is equivalent to dismiss the data records at that attribute, i.e., exclude the attribute from model input. $\square$

Algorithm 1 Interactive Bias Mitigation of Machine Classifiers

```

1: (for binary classification)
Input: Dataset  $D^{N \times M}$  having  $N$  records and  $M$  attributes  $X = \{x_1, x_2, \dots, x_M\}$  in each record.
2: Alice's target classifier  $\mathbb{F}$  on class label  $Y$ ; Bob's observation classifier  $\mathbb{F}_\emptyset$  on class label  $O$ .
Output: Transformation  $f$  for each identified bias-prone (bp) attribute:  $x_m \rightarrow x'_m$ , for  $\forall x_m \in X^{bp} \subseteq X$ .
3:
4: Initial  $X^{bp} = \emptyset$ .
5: while  $\max_{x_m \in X} d_\emptyset(\emptyset, x_m) > \epsilon$  do
6:   Step 1: Data Normalization
7:   for  $x_m \in X$  do
8:     if  $x_m$  is numerical attribute then
9:        $x_{m,i} \leftarrow (x_{m,i} - x_m^{\min}) / (x_m^{\max} - x_m^{\min})$   $\triangleright$  min-max normalization
10:
11:   Step 2: Distance Matrix Construction
12:   Compute attributes' contribution  $S_m$  to  $\mathbb{F}_\emptyset$ 's separating capacity.
13:   for  $x_a, x_b \in X, x_a \neq x_b$  do
14:     for  $h = 1 : H$  do
15:       Enumerate set  $X^*$  at level  $h$ .
16:       Compute  $w_{max}(\{X^*\}) = \sqrt{\frac{S_1^2 + S_2^2 + \dots + S_{|X^*|}^2}{|X^*|}}$ , for  $m = 1, 2, \dots, |X^*|$  for each  $X^*$ .
17:       Compute sub-distance  $d_{X^*}(x_a, x_b) = |w_{max}(\{X^*, x_a\}) - w_{max}(\{X^*, x_b\})|$  for each  $X^*$ .
18:       Compute overall distance  $d^H(x_a, x_b)$ .  $\triangleright$  distance matrix  $\mathbf{D}$ 
19:
20:   Step 3: Bias Concentration Determination
21:   Apply MDS to  $\mathbf{D}$ . Determine optimal dimension  $S$  from elbow plot.
22:   Determine coordinates  $o_{ms}$  of each attribute  $x_m$  at each bias dimension  $s$ .
23:   Determine bias concentration  $q_{ms}$ .  $\triangleright$  bias concentration matrix  $\mathbf{Q}$ 
24:   Indicate attributes' distance to origin  $d_\emptyset(\emptyset, x_m) = \sum_{s=1}^S (q_{sm} - q_{s\emptyset})^2$ , for  $m = 1, 2, \dots, M$ .
25:
26:   Step 4: Bias Mitigation
27:   Identify the attribute  $x_l$  having the largest distance to origin  $d_\emptyset(\emptyset, x_l)$ . Put  $x_l$  in  $X^{bp}$ .
28:   if  $x_l$  is numerical attribute then
29:     Search for  $\alpha$  s.t. after  $x_l \rightarrow x'_l = x_l^\alpha$  (repeat Step 1-3) there is  $d_\emptyset(\emptyset, x'_l) < \epsilon$ 
30:      $\triangleright$  polynomial transformation
31:   if  $x_l$  is categorical attribute then
32:     Search for combination of  $x_l$ 's categories s.t. after  $x_l \rightarrow x'_l$  (repeat Step 1-3) there is  $d_\emptyset(\emptyset, x'_l) < \epsilon$ 
33:      $\triangleright$  re-binning transformation
34:   Apply the determined transformation:  $x_l \rightarrow x'_l$ . Record the transformation  $f_{x_l}$ .
```

Overall, workflow of the framework is summarized in Algorithm 1. The algorithm consists of four steps: *Data normalization*, *Distance matrix construction*, *Bias concentration determination*, and *Bias mitigation*.

4 Case Study: UCI Adult Dataset

We apply our framework to the UCI adult dataset (Dua and Graff, 2019). This is a well-known dataset (e.g., Lee and Mangasarian, 2001; Caruana et al., 2004; Agrawal et al., 2005) for constructing the classifier to determine whether a person has an annual income over 50K USD. Recent studies reveal that this dataset entails gender discrimination in the classification (e.g., Calmon et al., 2017; Zhang et al., 2018; Chiappa, 2019; Sharma et al., 2019); the dataset thus serves as an important example for the existence of algorithmic bias, and is used as a benchmark for testing bias mitigation strategies.

Step 1: Data Normalization

The dataset contains $M = 14$ attributes, including 6 numerical attributes (*age*, *fnlwgt*, *education-num*, *capital-gain*, *capital-loss* and *hours-per-week*)¹⁵ and 8 categorical attributes (*workclass*, *education*, *marital status*, *occupation*, *relationship*, *race*, *sex* and *native country*). Class label Y is annual income being no less ($y = 1$) or less ($y = 0$) than 50K USD. We choose gender as the protected dimension O , removing *sex* from the attribute set X , which then has 13 elements. Dataset has $N = 48842$ records, divided into training set (32561 records) and test set (16281 records). Distribution of data records at the 13 attributes is shown in Figure 2.

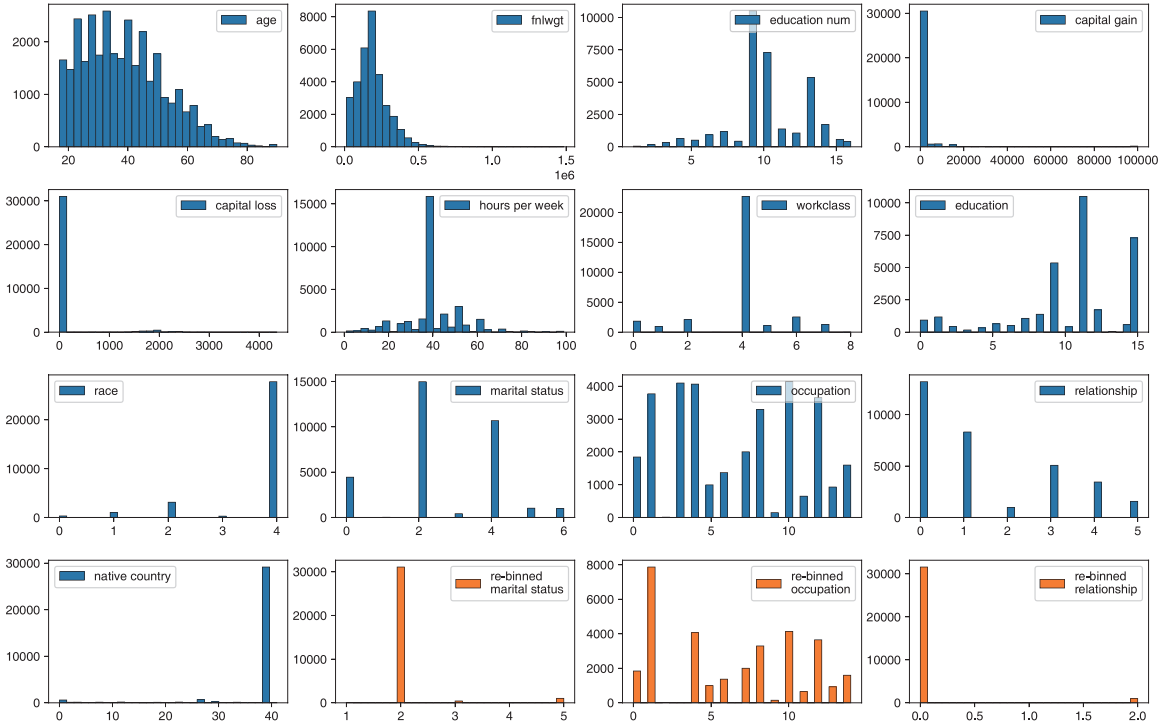


Figure 2: Distribution of data records at 13 attributes of the UCI adult dataset. For three attributes (*marital status*, *occupation*, *relationship*), distribution is changed after data transformation (Step 4, $\epsilon = 0.005$). Change in the distribution of *hours per week* after polynomial transformation is shown in Figure 6.

Values of original records span several orders of magnitude (Appendix C). Min-max normalization is applied on numerical attributes, placing data records within $[0, 1]$ interval (Table 2).

¹⁵Attribute *fnlwgt* (i.e., final weight) indicates how many people a recorded individual is representative of. It is not an intrinsic feature of the individual and some suggest this attribute to be removed during classification. We keep the attribute to reduce human engineering; tests suggest that results are similar when it is removed.

before normalization						
	age	fnlwgt	education num	capital gain	capital loss	hours per week
MIN	17	1.22E4	1	0	0	1
MAX	90	1.48E6	16	1.00E5	4356	99
AVE	38.58	1.90E5	10.08	1.08E3	87.30	40.44
STD	13.64	1.05E5	2.57	7.39E3	402.96	12.35

after normalization						
	age	fnlwgt	education num	capital gain	capital loss	hours per week
MIN	0	0	0	0	0	0
MAX	1	1	1	1	1	1
AVE	0.30	0.12	0.61	0.01	0.02	0.40
STD	0.19	0.07	0.17	0.07	0.09	0.13

Table 2: Statistics of data records at numerical attributes before/after min-max normalization.

Step 2: Distance Matrix Construction

For each pair of attributes (x_a, x_b) , we calculate the maximum separating capacity (equation (3)) at each option of companion attributes X^c for $|X^c| = 11$ till $|X^c| = 1$, and obtain their overall level-11 distance $d_D^{11}(x_a, x_b)$ with equation (8). The distance matrix is shown in Figure 3.

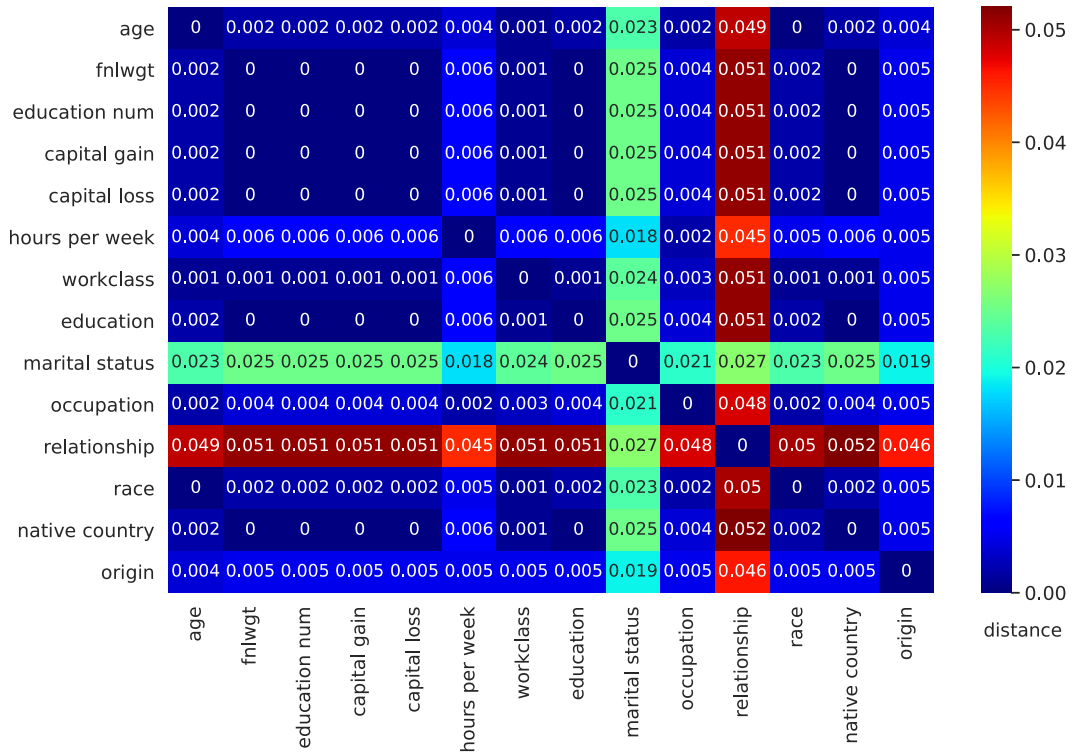


Figure 3: Distance matrix for attributes of the UCI adult dataset. Target Y is *income*; observable O is *sex*. The symmetric matrix records the pair-wise distance between the remaining 13 attributes plus the origin.

Relationship and *marital status* have the two largest (row) distance to origin and in general large distance to other attributes. Other attributes all have (row) distance to origin smaller than 0.005, clustering around the origin and around each other.

Step 3: Bias concentration determination

MDS analysis is conducted on the distance matrix to determine the appropriate number of bias sources with respect to the protected O . The elbow plot converges at $dim = 2$ (Figure 4a), indicating an appropriate cardinality $S = 2$ of bias sources. Correspondingly, attributes are projected onto the 2-D plane (Figure 4b).¹⁶ Attributes' updated distance to origin agree with the raw distance (Figure 3), where *relationship* and *marital status* float away from the cluster of other attributes. Bias concentration $Q^{S \times M} = Q^{2 \times 13}$ is indicated via the determined coordinates (Figure 4c).

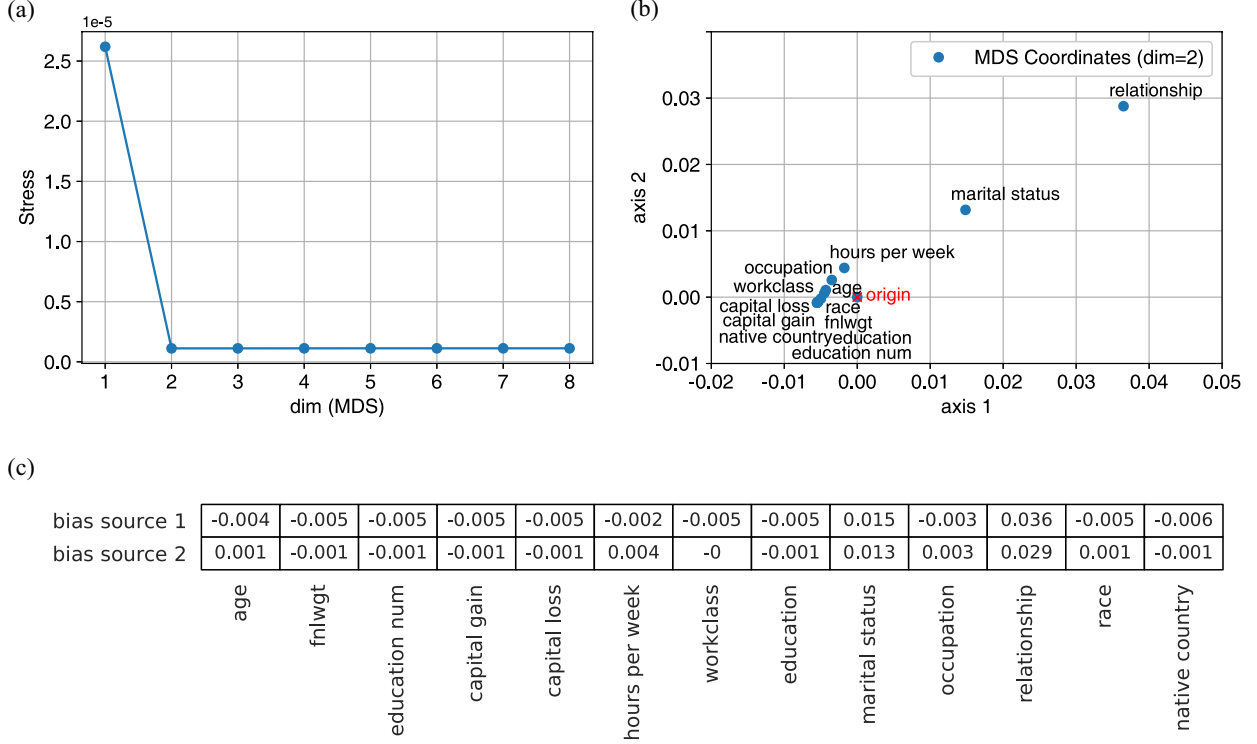


Figure 4: MDS on the distance matrix. (a) Elbow plot. Showing the appropriate dimension. (b) Projected 2-D attribute space. Showing attributes' distance to origin. (c) Bias concentration matrix. Showing attributes' coordinates at each identified bias source.

Step 4: Bias mitigation

Distance to origin indicates attributes' contribution to classifier $\mathbb{F}_O$'s separating capability, i.e., attributes' bias concentration with respect to the protected dimension O . We aim to restrict attributes' bias concentration within a threshold ϵ , i.e., $d_\theta(\emptyset, x_m) < \epsilon$. We choose two values ($\epsilon = 0.005, 0.01$) to show the results.¹⁷ We implement the greedy procedure to iteratively transform the most bias-prone attribute, until all attributes' distance to origin fall below ϵ . The iteration is shown in Table 4 and Figure 5. Searching for the appropriate transformation at each step takes different number of trials.

For $\epsilon = 0.01$, two iterations are needed. At Iteration #1, *relationship* is re-binned, merging four categories. *Relationship*'s distance to origin is suppressed (Figure 5a); other attributes' distance to origin are slightly affected (e.g., *marital status*'s distance to origin increases). At Iteration #2, *marital status* is re-binned, merging three categories. After the two steps, all attributes' distance to origin fall below $\epsilon = 0.01$.

¹⁶Coordinates determined by MDS preserve the distance between attributes, while maintain translational and rotational asymmetry. Translational asymmetry is eliminated as we set the origin of coordinates. Rotational asymmetry remains in the coordinates, but does not influence our results as we focus on attributes' distance instead of absolute coordinate.

¹⁷As mentioned, ϵ can be determined externally using the correlation of Y and O .

$\epsilon = 0.01$			
Iteration	Treated Attribute	Transformation	No. of Trials
#1	relationship	re-bin: Husband, Unmarried, Not-in-family, Wife	3
#2	marital status	re-bin: Married-civ-spouse, Never-married, Divorced	2

$\epsilon = 0.005$			
Iteration	Treated Attribute	Transformation	No. of Trials
#1	relationship	re-bin: Husband, Unmarried, Not-in-family, Wife, Own-child	4
#2	marital status	re-bin: Married-civ-spouse, Never-married, Divorced, Widowed	3
#3	hours per week	polynomial: x^5	2
#4	occupation	re-bin: Craft-repair, Adm-clerical	1

Table 3: Bias mitigation iteration at two values of ϵ (0.005, 0.01).

For $\epsilon = 0.005$, four iterations are needed due to the stricter condition. At Iteration #1, #2 and #4, *relationship*, *marital status* (not identical to the re-binning at $\epsilon = 0.01$) and *occupation* are re-binned, respectively. At Iteration #3, a polynomial transformation ($\alpha = 5$) applies to *hours per week*. After the four iterations, attributes' distance to origin fall below $\epsilon = 0.005$ (Figure 5b).

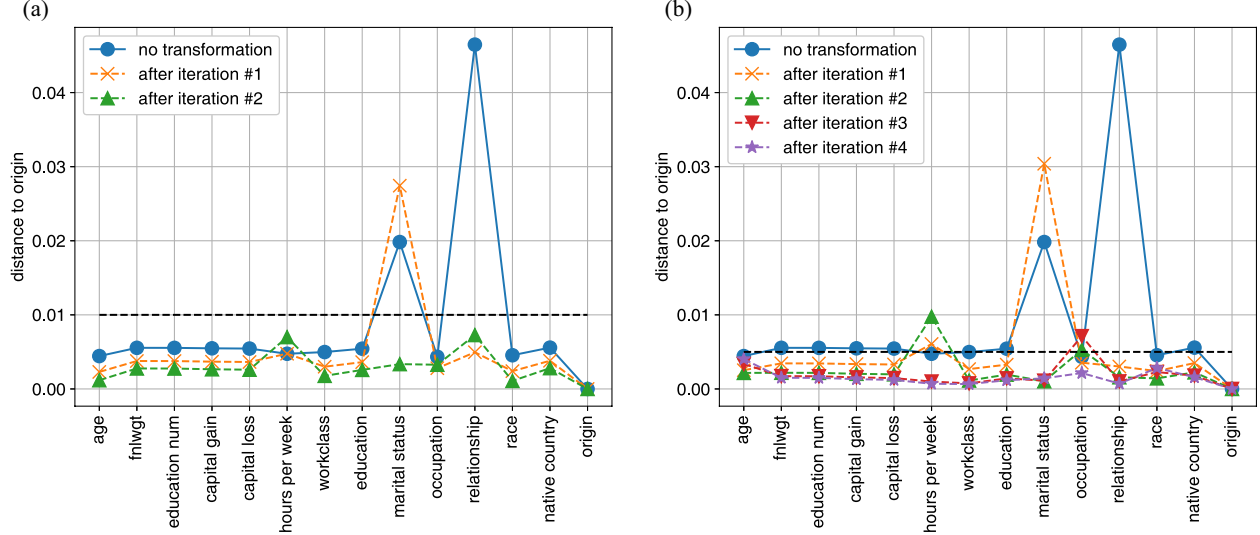


Figure 5: Attributes' distance to origin along iterations of bias mitigation. (a) $\epsilon = 0.01$. (b) $\epsilon = 0.005$.

As we desired, the determined transformations alter the distribution of data records. We use kernel density estimate (KDE) (Davis *et al.*, 2011) to visualize this change in distribution, which represents data with a continuous probability density curve in one or more dimensions.¹⁸ We show the result at $\epsilon = 0.005$. For the polynomial transformation at *hours per week*, KDE plot suggests that the absolute mean difference between (the min-max normalized) male and female records diminishes from 0.061 to 0.015 after the transformation. Bias concentration of *hours per week* at *O* (gender) is considerably reduced.

The change in distribution is further indicated with the Jensen–Shannon divergence (JS divergence) (Endres and Schindelin, 2003). For two distributions P_1, P_2 , $JS(P_1||P_2) = \frac{1}{2}KL(P_1||\frac{P_1+P_2}{2}) + \frac{1}{2}KL(P_2||\frac{P_1+P_2}{2})$, where the Kullback-Leibler divergence $KL(P_1||P_2) = \sum_{i=1}^n P_1(x_i) \log\left(\frac{P_1(x_i)}{P_2(x_i)}\right)$. We estimate the probability density function with a histogram having 20 bins; JS divergence diminishes from 0.039 to 0.012 after the polynomial transformation at *hours per week*.

¹⁸<https://seaborn.pydata.org/generated/seaborn.kdeplot.html>

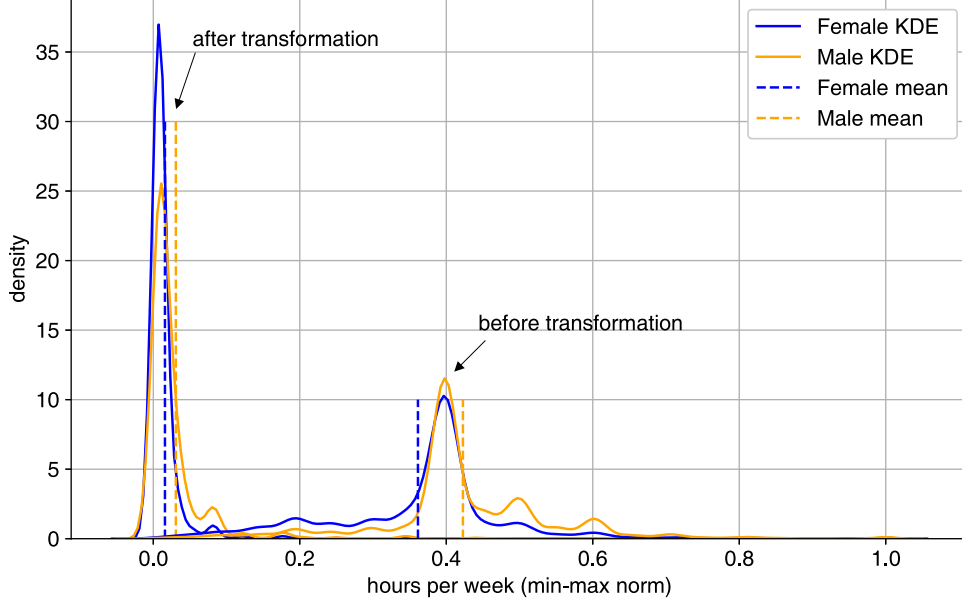


Figure 6: Effectiveness of polynomial transformation in reducing the bias concentration of *hours per week* at the two genders. Distribution of data records is visualized with the kernel density estimate (KDE) plot.

For re-binning transformation at the three attributes, change in their data distribution is shown in Figure 2.

Validation of data transformation

The determined transformations at bias-prone attributes are validated with a panel of choices for Alice’s classifier $\mathbb{F}$. We adopt 12 machine-learning models provided by the AutoML framework, Autogluon (Erickson et al., 2020): *LinearModel*, *KNeighborsUnif*, *KNeighborsDist*, *RandomForestGini*, *RandomForestEntr*, *ExtraTreesGini*, *ExtraTreesEntr*, *XGBoost*, *LightGBM*, *CatBoost*, *NeuralNetTorch*, *NeuralNetFastAI*, which are based on different algorithmic ideas (Table 4; see details in Appendix D). Training parameters are automatically optimized in Autogluon.

Model	Description
LinearModel	logistic regression classifier
KNeighborsUnif	k-nearest neighbors model (weight: uniform)
KNeighborsDist	k-nearest neighbors model (weight: distance)
RandomForestGini	random forest model (criterion: Gini Impurity)
RandomForestEntr	random forest model (criterion: information entropy)
ExtraTreesGini	extreme random forest model (criterion: Gini Impurity)
ExtraTreesEntr	extreme random forest model (criterion: information entropy)
XGBoost	XGBoost (GBDT) model: https://xgboost.readthedocs.io/
LightGBM	LightGBM (GBDT) model: https://lightgbm.readthedocs.io/
CatBoost	CatBoost (GBDT) model: https://catboost.ai/
NeuralNetTorch	PyTorch neural network models
NeuralNetFastAI	PyTorch neural network models with fastai backend

Table 4: Machine-learning models used in validating data transformation.

Our method does not rely on the specific measure of algorithm fairness; here we use *statistical parity* (e.g., Dwork et al., 2012) to indicate the fairness of classification outcomes. For this dataset, statistical parity asks that $\Pr_{\mathcal{D}}(\text{Income} \geq 50\text{K} \mid \text{sex} = \text{M}) = \Pr_{\mathcal{D}}(\text{Income} \geq 50\text{K} \mid \text{sex} = \text{F})$, where $\Pr_{\mathcal{D}}(\text{Income} \geq 50\text{K} \mid$

$\text{sex} = \text{M/F}$) is the conditional probability that the predicted annual income is no less than 50K given gender is male (M) or female (F). Fairness of model's classification outcome is thus indicated by the difference in probability, defined as

$$\Delta_{\text{SP}} = |\Pr_{\mathcal{D}}(\text{Income} \geq 50\text{K} \mid \text{sex} = \text{M}) - \Pr_{\mathcal{D}}(\text{Income} \geq 50\text{K} \mid \text{sex} = \text{F})|. \quad (13)$$

When the model's classification outcome is completely fair at this dimension (gender), $\Delta_{\text{SP}} = 0$; the larger the Δ_{SP} , the more unfair the classification.

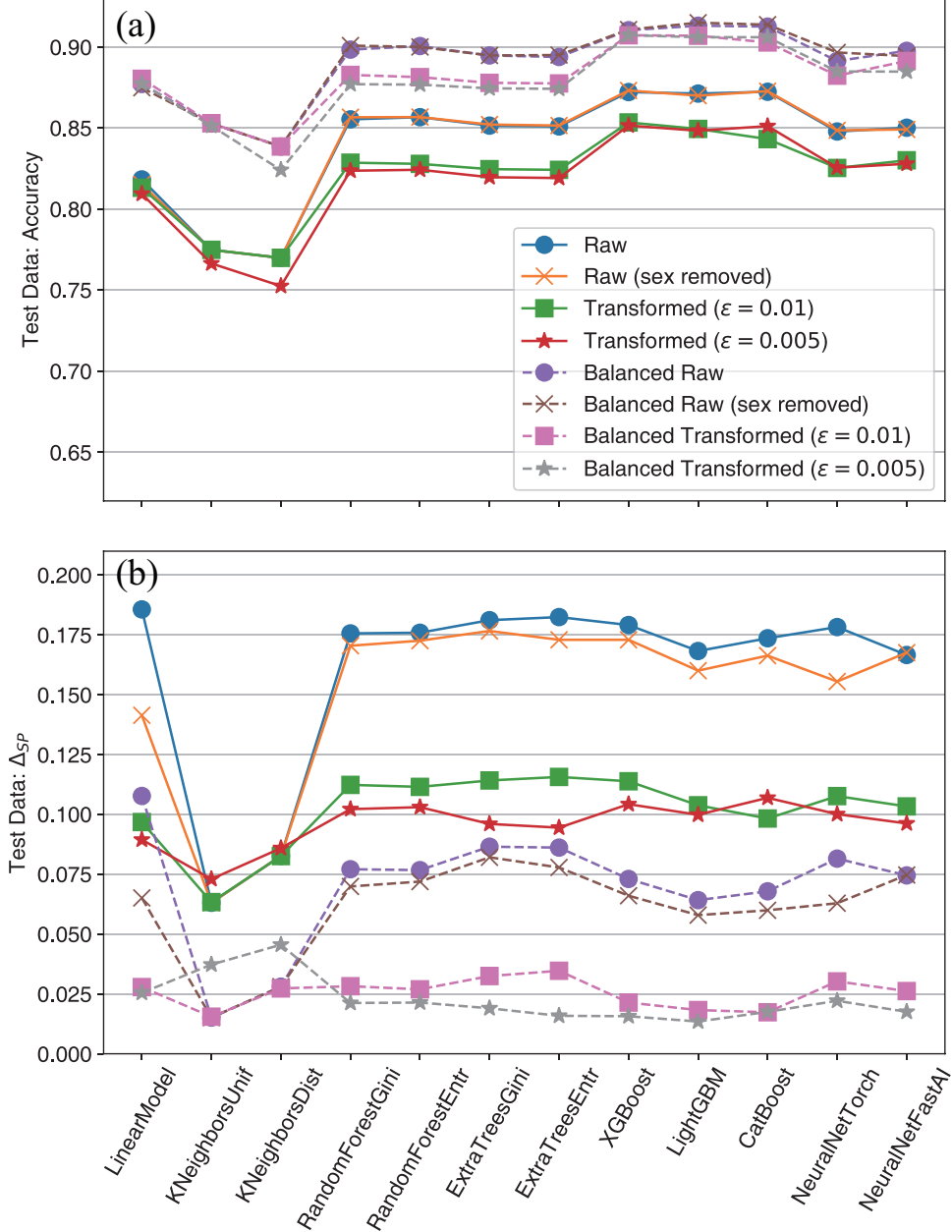


Figure 7: Classification (a) accuracy and (b) fairness across 12 classifiers. Comparing results before (the *sex* attribute included/removed) and after data transformation (at $\epsilon = 0.005$ and 0.01), and with unbalanced (solid lines) or balanced (dashed lines) test set.

We compare (i) classification accuracy on the test set and (ii) outcomes' statistical parity, before and after

the determined data transformation.

For the untransformed data, all 12 models well classify the test set, with an accuracy ranging from 75% to 91% (Figure 7a). There is a gap in the classification outcome at the two genders; an unsatisfactory fairness ($\Delta_{SP} > 0.15$) applies to 10/12 classifiers (Figure 7b). For the transformed data, there is a small reduction in classification accuracy (on average -2.3%). Desirably, classification’s fairness significantly improves, diminishing from an average value of $\Delta_{SP}^{Raw} = 0.159$, to $\Delta_{SP}^{Transformed} = 0.102$ and 0.096 for $\epsilon = 0.01$ and 0.005, respectively, overall an $> 37\%$ reduction. For further comparison, we consider the naive bias mitigation that only removes *sex* from the attribute set; gender bias persists after this treatment (comparing blue and yellow curves).

The $\Delta_{SP}^{Transformed}$ is still not close to zero. We find that it results from the unbalance of the test set: male have a higher probability of earning more than 50K a year if we randomly select a person from the raw test set (Table 4). We then construct a balanced test set by randomly selecting 590 out of 3256 male with high income and 4831 out of 7604 male with low income, and combining it with female’s data that have the same number in each category. Male and female will have an equal probability of having high income in this balanced test set. We re-run the classification and conduct the same comparison. With this balanced test set (dashed lines in Figure 7), classifiers achieve even higher accuracy, and the reduction of accuracy from data transformation is considerably reduced. The improvement in statistical parity is maintained across this treatment, and the $\Delta_{SP}^{Transformed}$ now approaches zero.

	Income $\geq$ 50 K	Income $<$ 50 K
male	3256	7604
female	590	4831

Table 5: Number of male/female in the test set having high/low income.

For 2/12 classifiers, our data transformation does not lead to clear improvement in fairness through sacrificing classification accuracy. This derives from that the *KNeighborsUnif* and *KNeighborsDist* classifiers are based on the k-nearest-neighbor (KNN) idea, where classification relies on records’ distance in the data space. As our transformation directly works on reshaping the data space and manipulating records’ distance, it undermines KNN classifiers’ training and leads to inferior results. Nevertheless, the small change across the transformation, in both classification accuracy and fairness, implies that our transformation does not drastically morph the generic distance structure of the data space.

5 Robustness Tests

We conduct the following robustness tests to validate the solution framework.

5.1 Complete distance matrix

As gender bias is identified as the dominant bias in this dataset, we check our result on this fact, by conducting MDS on the complete 15×15 distance matrix without *sex* excluded from the attribute set (Appendix E).

Consistent with the prior knowledge, results show that *sex* has the largest (raw) distance to origin among the 14 attributes (Figure A1a). This is the same for the updated distance to origin after MDS (Figure 8a), which determines $S = 2$ bias sources. *Sex*’s high *absolute* bias concentration (without O specified) is confirmed.

A rotation normalization $\mathbf{R}^{S \times S}$ is applied on $\mathbf{Q}^{S \times M}$, $S \times M = \mathbf{R}^{S \times S} \mathbf{Q}^{S \times M}$, such that attribute *sex* is projected as the first (since it is the strongest) bias source (B_1), i.e., $\tilde{q}_{B_1, \text{sex}} \neq 0$ and $\tilde{q}_{B_2, \text{sex}} = 0$. The first row of $\tilde{\mathbf{Q}}$ is then normalized by $\tilde{q}_{B_1, \text{sex}}$ and taking the absolute value, leading to the normalized bias concentration matrix $\tilde{\mathbf{Q}}$ (Figure 8b). The first row indicates how much each attribute contains gender bias B_1 . For example, the high concentration $q_{B_1, \text{relationship}}^{\text{norm}} = 0.47$ indicates that gender information is substantially encoded in the attribute *relationship*, revealing the considerable correlation between the classification on

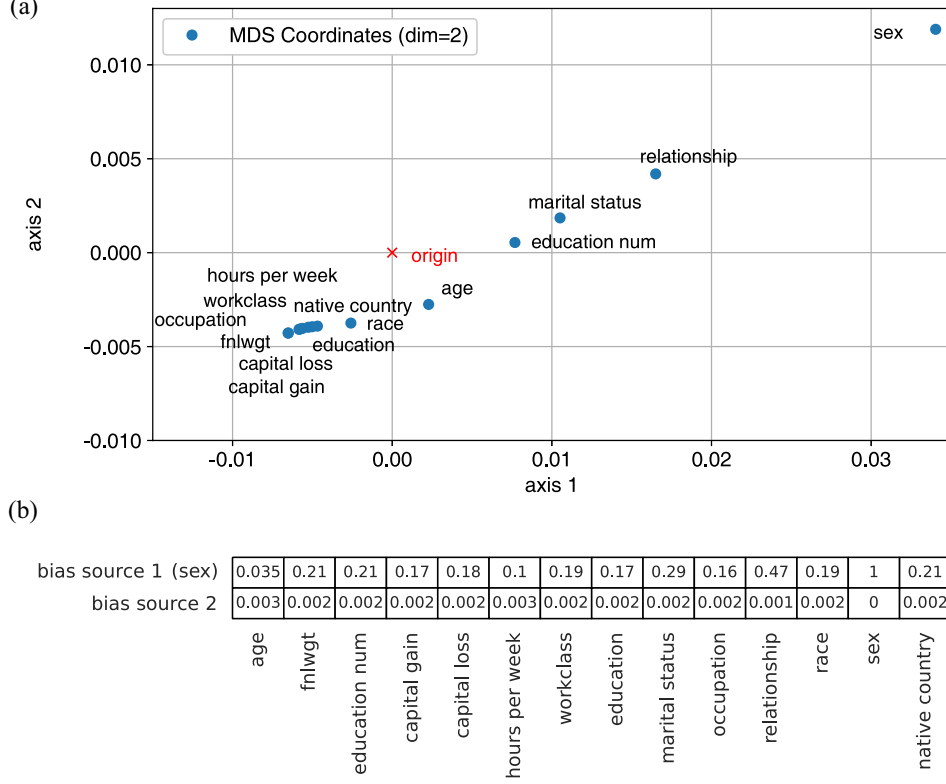


Figure 8: Identification of bias source on the complete attribute set without *sex* excluded. (a) Projected attribute space. (b) Rotation-normalized bias concentration matrix.

gender and the classification on relationship. This agrees with the main result where O is specified as gender, since relationship and marital status ($q_{B_1, \text{marital status}}^{\text{norm}} = 0.29$) as highlighted here are exactly the two most bias-prone attributes treated in the main result (Table 3).

5.2 Over-sampling the data space

We add synthetic records to the dataset (i.e., over-sampling) to test the robustness of attributes' distance to origin. We double the original 32561 data points; new data points fall within the min-max value range at each attribute (Table A2). We generate data using SMOTE (Synthetic Minority Over-sampling Technique) (Chawla *et al.*, 2002). A point z_{sample} and a point among its k nearest neighbors $z_{\text{sample}}^{\text{KNN}}$ are sampled, based on which a synthetic point $z_{\text{sample}}^{\text{new}} = z_{\text{sample}} + \gamma(z_{\text{sample}}^{\text{KNN}} - z_{\text{sample}})$ is created. We consider $\gamma \in [-0.1, 1.1]$ to create possible outliers outside the original value range. Result suggests that the distance measure remains robust in the over-sampled data space (Figure 9). We further generate completely random data points within the min-max range, ignoring data's structure across attributes; distance changes substantially this time, as data structure has been destroyed.

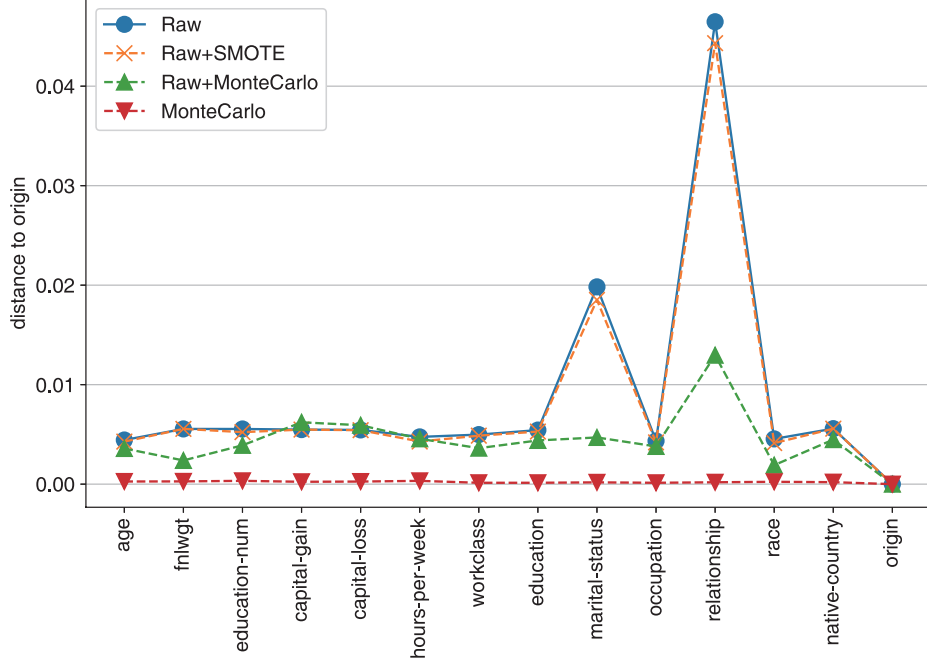


Figure 9: Over-sampling the data space. Results shown average over 100 stochastic runs.

5.3 Alternative transformation: noisification

We investigate a different idea of data transformation - adding noise to raw data (i.e., noisification) - besides our proposed transformation (i.e., polynomial/re-binning). For numerical attributes, we consider (1) a Gaussian noise multiplier n_g such that $z \rightarrow z' = z(1 + n_g)$, with $n_g \sim N(0, \sigma^2)$, (2) a Poisson noise at mean z , i.e., $z' = \text{Poisson}(z)$. For categorical attributes, we consider that there is a small probability p_n that each record is reassigned uniformly to any of its attribute's categories (including its current category).

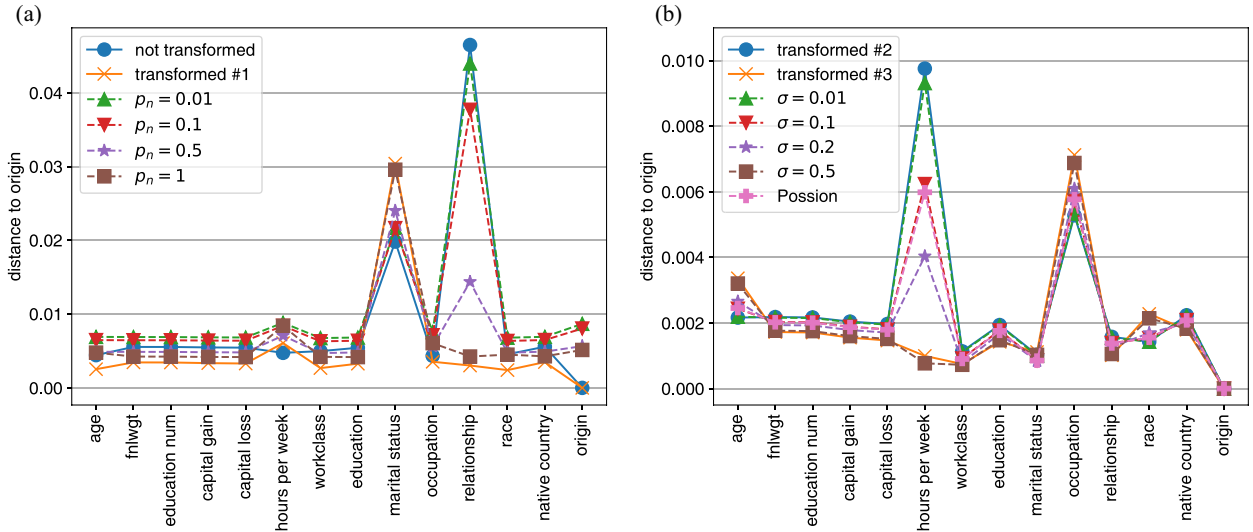


Figure 10: Noisification of data records. Results shown average over 100 stochastic runs. (a) Label-reassigning at categorical attribute *relationship*. (b) Gaussian/Poisson noise at numerical attribute *hours per week*.

We test the effectiveness of noisification at different σ and p_n . For categorical attribute *relationship* (Figure 10a), small reassigning probability does not bring down bias concentration; only when p_n is 0.5 or 1, does noisification’s effect resemble re-binning. Large p_n irreversibly destroys data records, however, while re-binning coherently preserves data structure. For numerical attribute *hours per week* (Figure 10b), similarly, only when noise is large ($\sigma = 0.5$ for Gaussian; Poisson noise not effective), does the effect resemble polynomial transformation, which preserves data structure better than noisification.

5.4 Alternative distance measure

We test an alternative expression for the overall distance measure $d_D^H(x_a, x_b)$ summarizing the sub-distance measures up to level H , following closely the expression of the SHAP value (Lundberg and Lee, 2017; Lundberg et al., 2020). In this case, sub-distance measures at each level are weighted in the sum (Appendix A), instead of unweighted (equation (8) and (9)).

Results under the weighted distance preserve the highlight of bias-prone attributes while emphasize the contrast in attributes’ distance to origin (Figure 11). Iteration order of bias mitigation (Table 4) is invariant across this change in distance measure. We vary the value of H for the two cases (weighted/unweighted sub-distance); results are robust at this parameter. We also check the result at $d = 3$, i.e., projecting attributes to a 3-D space; attributes’ distance to origin preserve the order.

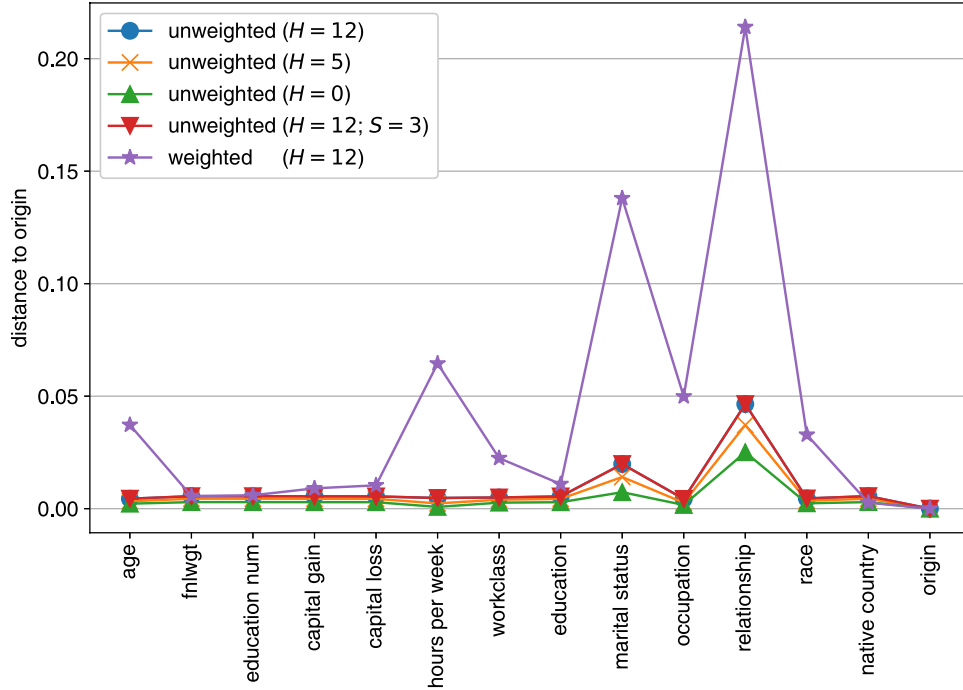


Figure 11: Attributes’ distance to origin at alternative settings (weighted/unweighted sub-distance, different values of H , 3-D attribute space).

5.5 Protecting the race attribute

Suppose Bob wishes to protect different racial groups instead of genders. This time *race* instead of *sex* is excluded from X , and the classification on O (*race*) is multi-nary. Distance measurement suggests that *sex*, *fmlwgt* and *education num* possess the largest bias concentration, whose distance to origin are furthest in the projected 3-D attribute space, as determined by MDS (Appendix F). Three iterations of transformation are needed to confine attributes’ bias concentration within $\epsilon = 0.01$: re-binning at *sex* (equivalently, remove this attribute as it only has two categories), polynomial transformation ($\alpha = 3$) at *fmlwgt*, and polynomial transformation ($\alpha = 1/7$) at *education num*.

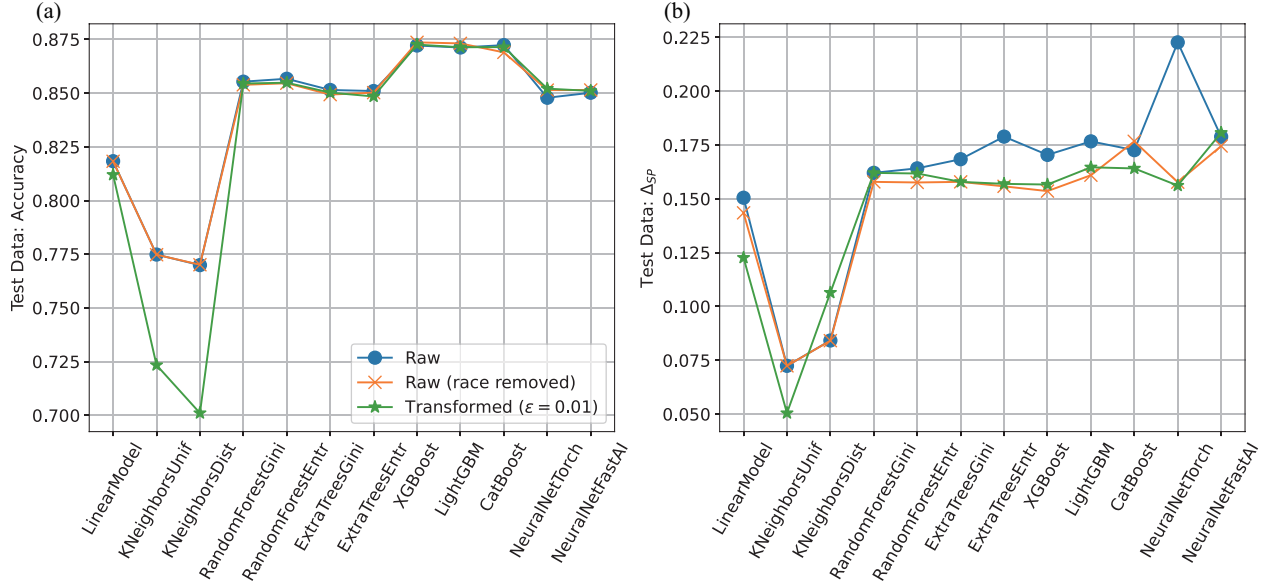


Figure 12: Performance of determined data transformation for protecting different racial groups. Classification (a) accuracy and (b) fairness across 12 classifiers. Comparing results before (*race* attribute included/removed) and after data transformation (at $\epsilon = 0.01$).²⁰

Across the 12 machine-learning models, the determined data transformation leads to slight reduction of classification accuracy, while increases classification fairness to certain extent (Figure 12).²¹ We further consider an alternative fairness metric, *equality of odds* (Hardt et al., 2016), to indicate classification fairness. Under this (arguably stricter (Putzel and Lee, 2022)) measure, the improvement of fairness is less concerted across the 12 models (Appendix F).

6 Case Study: Home Credit Default Risk Dataset

We further showcase our framework using the Home Credit Default Risk Dataset. This Kaggle competition dataset²² is used for constructing the classifier to predict Home Credit clients’ repayment abilities. Dataset has 120 attributes, including 72 numerical attributes and 48 categorical attributes (Appendix C). Class label Y is whether the client has repayment ability ($y = 1$) or not ($y = 0$). Similarly, we choose gender as the protected dimension O , removing *CODE GENDER* from the attribute set X . Dataset has $N = 307511$ records with gender information, including 202448 female, 105059 male and 4 unknown (discarded). A balanced test set having 61502 records is constructed by randomly selecting 2635 records with $y = 1$ and 28116 records with $y = 0$ for both the male and the female. The remaining 246005 records are used as the train set.

Data records are normalized and attributes’ distance matrix is constructed (consider $H = 1$; computational burden is large beyond $H = 2$ due to the enumeration of sub-distance).²³ MDS reveals that 2-D space is appropriate for projecting attributes; nine attributes are surfaced in the space (Figure 13a) as having more pronounced distance to origin than others.

²¹In this multi-nary classification scenario, fairness is indicated by the maximum pairwise difference between categories’ probability of statistically pairing. Extending the notion for binary classification, if the maximum probability of statistically pairing between any two categories is zero (thus all pairing probabilities are zero), the outcome achieves complete statistical parity.

²²<https://www.kaggle.com/competitions/home-credit-default-risk/data>. We use the ‘application train.csv’ dataset on the Kaggle main page.

²³Computation time increases drastically as H increases: for $H = 0, 1, 2$, the program finishes in 10 seconds, 20 minutes and 19 hours, respectively.

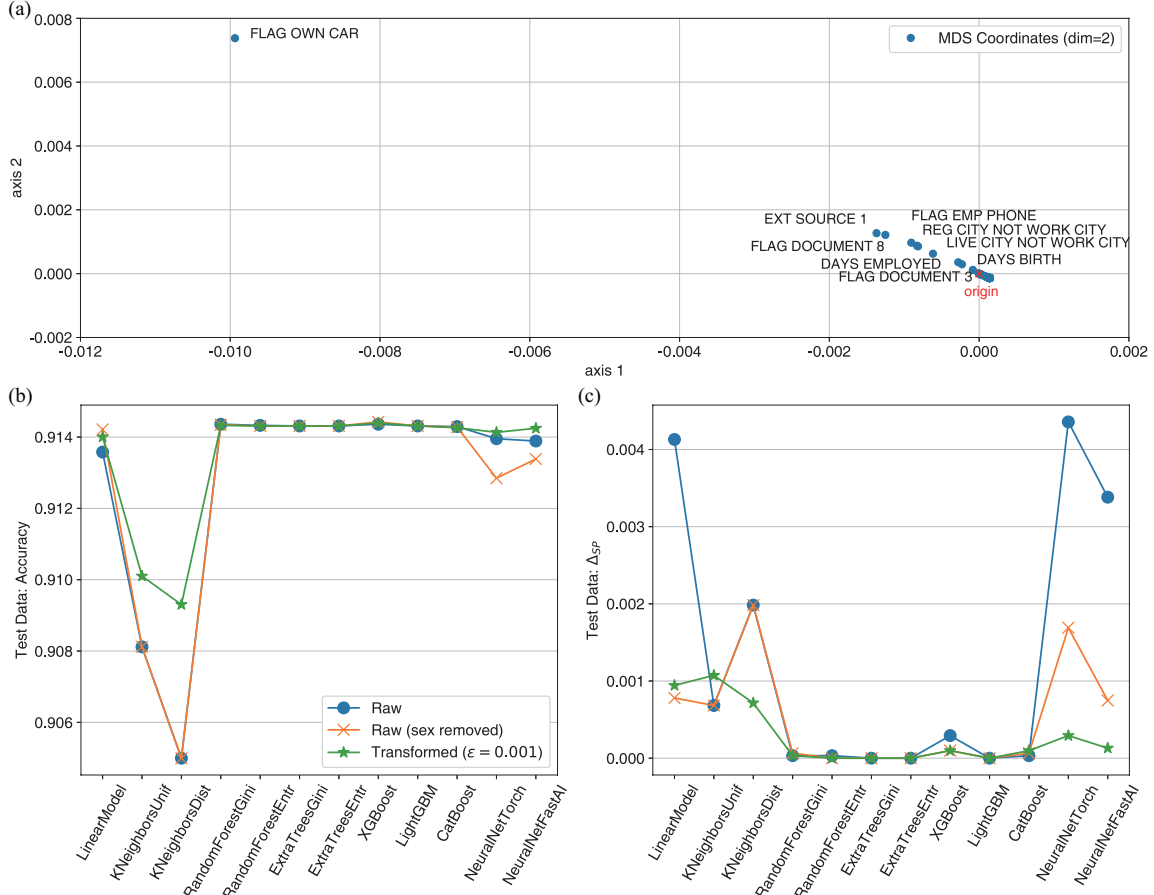


Figure 13: Bias mitigation at the Home Credit Default Risk Dataset. Target Y is repayment ability; observable O is gender. (a) Projected attribute space. (b) Classification accuracy. (c) Classification fairness.

Ten iterations are needed to gradually suppress attributes’ maximum bias concentration below $\epsilon = 0.001$ (Table 6). The determined data transformations perform well on the 12 machine-learning models (Figure 13b, c). In 7/12 cases, the transformation only leads to slight change in the result; at the other 5 classifiers (*LinearModel*, *KNeighborsUnif*, *KNeighborsDist*, *NeuralNetTorch*, *NeuralNetFastAI*), the transformation achieves higher classification accuracy and enhanced classification fairness (except for one case at *KNeighborsUnif*, a model sensitive to our data transformation, as mentioned) at the same time.

The change in classification accuracy and fairness is not substantial in absolute number, however, as models have already achieved good performance (accuracy $> 90\%$, $\Delta_{SP} < 0.5\%$) before transformation. The small Δ_{SP} across models suggests that this dataset does not embed substantial gender bias in the first place.²⁴ This result nevertheless demonstrates a valuable example of completely overriding the fairness-utility tradeoff, at a specific utility measure (accuracy) and a particular fairness measure (statistical parity).

²⁴When checking the distances on the original attribute set without *CODE GENDER* excluded, *CODE GENDER* only has the 13th largest raw distance to origin, confirming the small gender bias concentration of this dataset. This is the very reason that we can realise a small ϵ in this case.

$\epsilon = 0.001$

Iteration	Treated Attribute	Transformation	No. of Trials
#1	FLAG OWN CAR	re-bin: $2 \rightarrow 1$	1
#2	EXT SOURCE 1	polynomial: x^7	3
#3	FLAG DOCUMENT 8	re-bin: $2 \rightarrow 1$	1
#4	FLAG EMP PHONE	re-bin: $2 \rightarrow 1$	1
#5	REG CITY NOT WORK CITY	re-bin: $2 \rightarrow 1$	1
#6	DAYS EMPLOYED	polynomial: $x^{1/3}$	1
#7	LIVE CITY NOT WORK CITY	re-bin: $2 \rightarrow 1$	1
#8	FLAG DOCUMENT 3	re-bin: $2 \rightarrow 1$	1
#9	DAYS BIRTH	polynomial: x^5	2
#10	FLAG DOCUMENT 6	re-bin: $2 \rightarrow 1$	1

Table 6: Bias mitigation iteration at the Home Credit Default Risk Dataset. For re-binning, showing the change in number of categories across transformation.

7 Concluding Remarks

We developed a framework for identification, projection and mitigation of algorithmic bias. We consider a *propose and review* scenario between two stakeholders: Alice proposes to utilize data records for constructing machine classifier $\mathbb{F}$ at outcome Y , then Bob reviews the proposal to protect a sensitive dimension O from being discriminated on. Bob identifies potential source of discrimination with respect to O by constructing an observing classifier $\mathbb{F}_O$. Separating capability of $\mathbb{F}_O$ corresponds to input attributes’ bias concentration, which Bob aims to mitigate. He measures attributes’ contribution to $\mathbb{F}_O$ ’s separating capability and identifies sources of bias in the projected attribute space. By iteratively applying appropriate transformation to data records at the most bias-prone attribute, data’s bias concentration with respect to the projected O is suppressed in a greedy manner. Data transformations are approved by Bob and are transferred to Alice’s target $\mathbb{F}$.

The method achieves good performance at the UCI adult dataset. Determined transformations substantially reduced the unfairness of classification outcome, while largely preserved classification accuracy, across 10/12 machine-learning models based on different algorithmic ideas. Result confirms previous findings that gender bias permeates this dataset, and is robust against over-sampling the data space, noisification as alternative transformation, alternative distance measure, and different choices of O . Method performs consistently at the larger, HCDR dataset. Classification fairness is enhanced at 5/12 models and unchanged at the rest models but one, with classification accuracy unchanged or even improved across the 12 models.

A few design ideas are highlighted in this study:

- We do not assume prior knowledge of bias sources/attributes, while consider the general case of zero knowledge of bias existence and bias embedding. Bias sources are inferred from data.
- Our method does not rely on metrics of classification outcome’s fairness. Fairness is evaluated in the projected bias concentration space, and is achieved via confining attribute’s distance to origin.
- We consider different families of transformation for the two types of attributes (categorical, numerical). We specify the algorithmic order for finding the appropriate transformation in each case.
- Fundamentally, the fairness-utility tradeoff is obviated by separating two classifiers: reducing bias concentration at the observing classifier while preserving the separability of the target classifier. We document an unequivocal case where fairness and utility under certain definitions are both enhanced, completely overriding the tradeoff.

Limitations and future directions

Our study is limited in a number of ways, and opens avenues for future research.

Besides using centroid of data records to characterize dataset’s separability w_{max} , we can investigate alternative measures. For example, one well-adopted approach considers using support vector machines (SVM) for $\mathbb{F}_0$, and the problem concerns the separating hyperplanes in SVM’s model space. Hyperplane separation theorem (Boyd et al., 2004) determines that there always exists the separating hyperplane if dataset can be allocated to disjoint convex subsets. The maximum separating capability of a dataset can then be indicated by the SVM’s largest separating margin.

The framework focuses on binary classification, and the extension to multi-nary classification welcomes further investigation. Such extension concerns characterizing attributes’ separating capacity w_{max} (equation (5)), and selecting appropriate metric to indicate improvement in classification’s fairness (Note 19). As shown in the results (Section 5.5 and Appendix F), our method may not demonstrate amelioration of discrimination under particular fairness metrics.

We consider a restricted family of transformation for each type of attributes. This leaves substantial space for future study. On the one hand, categorization of attributes can be more granular (e.g., ordinal categorical, non-ordinal categorical (Appendix B), mixed etc); on the other hand, alternative mathematical forms can be employed. Polynomial/re-binning transformations preserve data structure to certain extent; alternative transformations should not drastically destroy data structure, as in noisification (Section 5.3).

The method’s scalability is imperfect. Computational obstacles lie in the enumeration of sub-distance measures, in the uncertainty around MDS result, and in the iteration during bias mitigation (where both convergence and effectiveness are not guaranteed, and finding the appropriate transformation may take many trials at one iteration). For large datasets having massive records and abundant attributes, the computation could be daunting. This computational burden has a positive side, however, as it is then difficult for Alice to infer the dimension Bob aims to protect, preventing reverse-engineering of the framework.

Managerial implications

The usage scenario of this framework is not restricted to between regulator and enterprise. More generally, it applies to a pair of data provider and data utilizer. Similar to public agencies, private data providers concern that they would be responsible for the discriminatory issues induced by machine classifiers that utilize their data. Reviewing data’s use case and conducting bias mitigation well align with data providers’ interests. More over, a data owner (e.g., a platform that generates and collects data) may ask a data analyzer (e.g., a consulting IT company that sells algorithms as products) to build a machine classifier for the owner itself, during which the data owner may desire to impose constraints on the product’s classification capability at certain dimensions. These application landscapes extend the current regulator-enterprise scenario.

On the reviewer’s side, the framework can be used as a tool for proof-checking bias concentration at a dataset’s existing attributes, if the reviewer does not have a protected dimension in mind but has access to Alice’s classifier. That is, let $O = Y$, $\mathbb{F}_0 = \mathbb{F}$. In this (arguably more common) case, attributes’ distance to origin indicate their contribution to model’s classification capability, and certain signal can be determined as bias to be removed, directly hitting the fairness-utility tradeoff. This is a particular case of the general propose and review scenario, where Bob and Alice stand on the same side, without private information preserved at both.

Taking a step back, however, the utility of the framework relies on its adoption in the first place. In fact, adoption is a core behavioral issue for algorithmic bias mitigation in general. It is found that, even if bias-mitigating algorithms are shown to be effective in mitigating racial inequality, the low adoption of these algorithms among minority groups limits their positive effects (Zhang et al., 2021). Unlike adoption of algorithms in decision-making, which is driven by apparent economic incentives and is largely spontaneous among organizations, adoption of algorithmic bias mitigation may conflict with current commercial practice, calling for the advent of new business models and strategic incentivization campaigns.

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Appendix

Appendix A: Alternative distance measure

We consider an alternative expression for the overall distance measure $d_D^H(x_a, x_b)$ summarizing the sub-distance measures up to level H , following closely the expression of the SHAP value (*Lundberg and Lee, 2017; Lundberg et al., 2020*). In this case, sub-distance measures at each level are weighted in the sum, instead of unweighted (equation (8) and (9)).

We have

$$\begin{aligned} d_D^H(x_a, x_b) \triangleq & \frac{1}{C_2^{|X|} C_0^{|X|-2}} d_{X/\{x_a, x_b\}} + \sum_{x_m^1 \in X/\{x_a, x_b\}} \frac{1}{C_2^{|X|} C_1^{|X|-2}} d_{X/\{x_a, x_b, x_m^1\}} + \dots \\ & + \sum_{\{x_{m_1}^H, \dots, x_{m_H}^H\} \in X/\{x_a, x_b\}} \frac{1}{C_2^{|X|} C_H^{|X|-2}} d_{X/\{x_a, x_b, x_{m_1}^H, \dots, x_{m_H}^H\}}. \end{aligned} \quad (14)$$

For attributes' (raw) distance to origin, there is

$$\begin{aligned} d_D^H(x_a, \emptyset) \triangleq & \frac{1}{C_1^{|X|} C_0^{|X|-1}} d_{X/\{x_a\}} + \sum_{x_m^1 \in X/\{x_a\}} \frac{1}{C_1^{|X|} C_1^{|X|-1}} d_{X/\{x_a, x_m^1\}} + \dots \\ & + \sum_{\{x_{m_1}^H, \dots, x_{m_H}^H\} \in X/\{x_a\}} \frac{1}{C_1^{|X|} C_H^{|X|-1}} d_{X/\{x_a, x_{m_1}^H, \dots, x_{m_H}^H\}}, \end{aligned} \quad (15)$$

where

$$d_{X/\{x_a, x_{m_1}^H, \dots, x_{m_H}^H\}} = |w_{max}(\{X/\{x_a, x_{m_1}^H, \dots, x_{m_H}^H\}\}) - w_{max}(\{X/\{x_{m_1}^H, \dots, x_{m_H}^H\}\})|. \quad (16)$$

The number of terms in $d_D^H(x_a, x_b)$ and $d_D^H(x_a, \emptyset)$ is the same as before, despite the prolonged expression of the sum.

Appendix B: Example of re-binning categorical attributes

Among categorical attributes, we distinguish ordinal attributes where the categories have a particular order (e.g., age groups: < 30 , $30 - 40$, $40 - 50$ etc), from non-ordinal attributes where the categories do not have an order (e.g., marital status: single, married, divorced, etc).

- For ordinal attributes, re-binning only combines neighboring categories. Suppose the original data records fall into R ordinal categories. There are C_{r-1}^{R-1} options for re-binning the R ordinal bins into r bins. As r spans from 1 to R , the total number of re-binning options is

$$C_0^{R-1} + C_1^{R-1} + \dots + C_{R-1}^{R-1} = 2^{R-1}. \quad (17)$$

Subtracting the two plain options ($r = 1$ or R), there are $2^{R-1}-2$ scenarios of re-binning.

In search for the appropriate re-binning such that the maximum bias concentration condition is met, we start from $r = R - 1$, i.e., merging two bins into one, and decrease r until $r = 2$, i.e., only two bins remain.

- For non-ordinal attributes, re-binning can combine non-neighboring categories. Suppose the original data records fall into R non-ordinal categories. The number of possible partitions of the R bins is given by the Bell numbers ²⁵, B_R , which satisfies the recurrence relation:

$$B_{R+1} = \sum_{r=0}^R C_r^R B_r; \quad B_0 = B_1 = 1. \quad (18)$$

Similarly, there are $B_R - 2$ scenarios of re-binning.

As the categories have no order, we implement a different search order in finding the appropriate re-binning for non-ordinal attributes. In order to reduce the difference between the frequency of positive and negative class (as this difference indicates data's separation), at each step, the two (possibly already re-binned) categories having the largest positive and the smallest negative frequency difference, are re-binned into one new category. The search ends when the maximum bias concentration condition is met.

For example, the attribute *relationship* in the UCI adult dataset has 6 non-ordinal categories (Table A1), each having specific frequency difference between the positive and the negative class. At the first trial of re-binning, the two categories (Husband, Unmarried) are re-binned into one new category, as they have the largest positive and smallest negative frequency difference, respectively. ²⁶

	Husband	Not-in-family	Other-relative	Own-child	Unmarried	Wife
No. Male	13192	4430	551	2823	792	2
No. Female	1	3875	430	2245	2654	1566
Frequency Difference	0.605	-0.156	-0.015	-0.079	-0.210	-0.145

Table A1: Frequency difference between positive/negative class for the non-ordinal attribute *relationship*).

²⁵https://en.wikipedia.org/wiki/Bell_number

²⁶As mentioned in the main text (equation (4)), frequency difference S_m for attribute x_m is calculated as follows. For example, here we have $N^+ = 13192 + 4430 + 551 + 2823 + 792 + 2 = 21790$ and $N^- = 1 + 3875 + 430 + 2245 + 2654 + 1566 = 10771$. For the category *Husband*, its frequency at male is $13192/21790$, at female is $1/10771$ (this is a demonstration of gender information embodied in the *relationship* attribute; the one case at female may correspond to a female being a lesbian but playing the role as husband in the relationship), and the frequency difference is $13192/21790 - 1/10771 = 0.605$.

Appendix C: Data information

UCI adult dataset

	age	fnlwgt	education num	capital gain	capital loss	hours per week	workclass
MIN	17	1.22E4	1	0	0	1	0
MAX	90	1.48E6	16	1.00E5	4356	99	8
AVE	38.58	1.90E5	10.08	1.08E3	87.30	40.44	-
STD	13.64	1.05E5	2.57	7.39E3	402.96	12.35	-
	education	marital status	occupation	relationship	race	sex	native country
MIN	0	0	0	0	0	0	0
MAX	15	6	14	5	4	1	41

Table A2: Statistics of the UCI adult dataset (integers at categorical attributes indicate category indices).

HCDR dataset

	NAME CONTRACT TYPE	CODE GENDER	FLAG OWN CAR	FLAG OWN REALTY	NAME TYPE SUITE	NAME INCOME TYPE	NAME EDUCATION TYPE	NAME FAMILY STATUS
MIN	0	0	0	0	0	0	0	0
MAX	1	1	1	1	7	7	4	5
	NAME HOUSING TYPE	FLAG MOBIL	FLAG EMP PHONE	FLAG WORK PHONE	FLAG CONT MOBILE	FLAG PHONE	FLAG EMAIL	OCCUPATION TYPE
MIN	0	0	0	0	0	0	0	0
MAX	5	1	1	1	1	1	1	18
	WEEKDAY START APPR PROCESS START	REG REGION NOT LIVE REGION	REG REGION NOT WORK REGION	LIVE REGION NOT WORK REGION	REG CITY NOT LIVE CITY	REG CITY NOT WORK CITY	LIVE CITY NOT WORK CITY	ORGANIZATION TYPE
MIN	0	0	0	0	0	0	0	0
MAX	6	1	1	1	1	1	1	57
	FONDKA- PREMONT MODE	HOUSE- TYPE MODE	WALLS- MATERIAL MODE	EMERGENCY- STATE MODE	FLAG DOCUMENT 2	FLAG DOCUMENT 3	FLAG DOCUMENT 4	FLAG DOCUMENT 5
MIN	0	0	0	0	0	0	0	0
MAX	4	3	7	2	1	1	1	1
	FLAG DOCUMENT 6	FLAG DOCUMENT 7	FLAG DOCUMENT 8	FLAG DOCUMENT 9	FLAG DOCUMENT 10	FLAG DOCUMENT 11	FLAG DOCUMENT 12	FLAG DOCUMENT 13
MIN	0	0	0	0	0	0	0	0
MAX	1	1	1	1	1	1	1	1
	FLAG DOCUMENT 14	FLAG DOCUMENT 15	FLAG DOCUMENT 16	FLAG DOCUMENT 17	FLAG DOCUMENT 18	FLAG DOCUMENT 19	FLAG DOCUMENT 20	FLAG DOCUMENT 21
MIN	0	0	0	0	0	0	0	0
MAX	1	1	1	1	1	1	1	1

Table A3: Statistics of the HCDR dataset (categorical attributes).

	CNT CHILDREN	AMT INCOME	AMT CREDIT	AMT ANNUITY	AMT GOODS	REGION POPULATION RELATIVE	DAYS BIRTH	DAYS EMPLOYED
MIN	0.00	2.56E4	4.50E4	1.62E3	4.05E4	2.90E-4	-2.52E4	-1.79E4
MAX	19.00	1.17E8	4.05E6	2.58E5	4.05E6	0.07	-7.49E3	3.65E5
AVE	0.41	1.67E5	5.99E5	2.70E4	5.38E5	0.02	-1.61E4	6.56E4
STD	0.72	2.59E5	4.02E5	1.44E4	3.69E5	0.01	4.37E3	1.43E5
	DAYS ID REGISTRATION	DAYS PUBLISH	OWN CAR AGE	CNT FAM MEMBERS	REGION RATING CLIENT	REGION RATING RATING CLIENT W CITY	HOURL APPR PROCESS START	EXT SOURCE 1
MIN	-2.34E4	-7.20E3	0.00	1.00	1.00	1.00	0.00	0.01
MAX	0.00	0.00	91.00	20.00	3.00	3.00	23.00	0.96
AVE	-5.01E3	-3.00E3	12.05	2.15	2.05	2.03	12.06	0.51
STD	3.54E3	1.51E3	12.04	0.91	0.51	0.50	3.27	0.21
	EXT SOURCE 2	EXT SOURCE 3	APARTMENTS AVG	BASEMENT- AREA AVG	YEARS BEGIN- EXPLUATATION AVG	YEARS BUILD AVG	COMMON- AREA AVG	ELEVATORS AVG
MIN	1.32E-6	5.27E-4	0.00	0.00	0.00	0.00	0.00	0.00
MAX	0.85	0.89	1.00	1.00	1.00	1.00	1.00	1.00
AVE	0.51	0.51	0.12	0.09	0.98	0.75	0.04	0.08
STD	0.19	0.19	0.11	0.08	0.06	0.11	0.08	0.13
	ENTRANCES AVG	FLOORSMAX AVG	FLOORSMIN AVG	LANDAREA AVG	LIVING- APARTMENTS AVG	LIVINGAREA AVG	NONLIVING- APARTMENTS AVG	NONLIVING- AREA AVG
MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MAX	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
AVE	0.15	0.23	0.23	0.07	0.10	0.11	0.01	0.03
STD	0.10	0.14	0.16	0.08	0.09	0.11	0.05	0.07
	APARTMENTS MODE	BASEMENT- AREA MODE	YEARS BEGIN- EXPLUATATION MODE	YEARS BUILD MODE	COMMONAREA MODE	ELEVATORS MODE	ENTRANCES MODE	FLOORSMAX MODE
MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MAX	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
AVE	0.11	0.09	0.98	0.76	0.04	0.07	0.15	0.22
STD	0.11	0.08	0.06	0.11	0.07	0.13	0.10	0.14
	FLOORSMIN MODE	LANDAREA MODE	LIVING- APARTMENTS MODE	LIVING- AREA MODE	NONLIVING- APARTMENTS MODE	NONLIVING- AREA MODE	APARTMENTS MEDI	BASEMENT- AREA MEDI
MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MAX	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
AVE	0.23	0.07	0.11	0.11	0.01	0.03	0.12	0.09
STD	0.16	0.08	0.10	0.11	0.05	0.07	0.11	0.08
	YEARS BEGIN- EXPLUATATION MEDI	YEARS BUILD MEDI	COMMON- AREA MEDI	ELEVATORS MEDI	ENTRANCES MEDI	FLOORSMAX MEDI	FLOORSMIN MEDI	LANDAREA MEDI
MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MAX	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
AVE	0.98	0.76	0.04	0.08	0.15	0.23	0.23	0.07
STD	0.06	0.11	0.08	0.13	0.10	0.14	0.16	0.08
	LIVING- APARTMENTS MEDI	LIVING- AREA MEDI	NONLIVING- APARTMENTS MEDI	NONLIVING- AREA MEDI	TOTAL- AREA MODE	OBS 30 CNT SOCIAL CIRCLE	DEF 30 CNT SOCIAL CIRCLE	OBS 60 CNT SOCIAL CIRCLE
MIN	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
MAX	1.00	1.00	1.00	1.00	1.00	47.00	7.00	47.00
AVE	0.10	0.11	0.01	0.03	0.10	1.42	0.14	1.41
STD	0.09	0.11	0.05	0.07	0.11	2.32	0.44	2.30
	DEF 60 CNT SOCIAL CIRCLE	DAYS LAST PHONE CHANGE	AMT REQ CREDIT BUREAU HOUR	AMT REQ CREDIT BUREAU DAY	AMT REQ CREDIT BUREAU WEEK	AMT REQ CREDIT BUREAU MON	AMT REQ CREDIT BUREAU QRT	AMT REQ CREDIT BUREAU YEAR
MIN	0.00	-4.18E3	0.00	0.00	0.00	0.00	0.00	0.00
MAX	7.00	0.00	4.00	6.00	8.00	27.00	261.00	22.00
AVE	0.10	-963.54	0.01	0.01	0.03	0.27	0.27	1.90
STD	0.36	825.91	0.08	0.11	0.21	0.92	0.83	1.87

Table A4: Statistics of the HCDR dataset (numerical attributes).

Appendix D: Machine-learning models

Autogluon (*Erickson et al.*, 2020) is a framework for AutoML, where the parameters of machine-learning models are automatically optimized. We use the following models in Autogluon, some of which refer to the scikit-learn toolbox (*Pedregosa et al.*, 2011).

LinearModel is the logistic regression model in scikit-learn. Logistic regression is also known as logit regression, maximum-entropy classification or the log-linear classifier. In this model, the probabilities describing the possible outcomes of a single trial are modeled using a logistic function.

KNeighborsUnif and **KNeighborsDist** are two variants (parameter *weight* = ‘uniform’/‘distance’) under the *KNeighborsClassifier* in scikit-learn. *KNeighborsClassifier* implements learning based on the k nearest neighbors of each data point, with integer k specified by the user. Classification is computed from a majority vote of the k neighbors. At uniform weighting, all points in each neighborhood are weighted equally. At distance weighting, points are weighted by the inverse of their distance; closer neighbors of a query point have a greater influence than distant neighbors.

RandomForestGini and **RandomForestEntr** are two variants (parameter *criterion* = ‘gini’/‘entropy’) under the *RandomForestClassifier* in scikit-learn. A random forest is a meta estimator that fits a number of decision tree classifiers on various sub-samples of the dataset and uses averaging to improve the predictive accuracy and control over-fitting. The criterion to measure the quality of a split is Gini impurity at ‘gini’ and Shannon information entropy at ‘entropy’.

ExtraTreesGini and **ExtraTreesEntr** are two variants (parameter *criterion* = ‘gini’/‘entropy’) under the *ExtraTreesClassifier* in scikit-learn, i.e., extremely randomized trees classifier. As in random forests, a random subset of candidate features are used, but instead of looking for the most discriminative thresholds, thresholds are drawn at random for each candidate feature and the best of these randomly-generated thresholds is picked as the splitting rule.

XGBoost is an optimized distributed library (*Chen and Guestrin*, 2016) that implements machine learning algorithms under the Gradient Boosting framework. A Gradient Boosting Decision Trees (GBDT) is a decision tree ensemble learning algorithm similar to random forest. GBDTs iteratively train an ensemble of shallow decision trees, with each iteration using the error residuals of the previous model to fit the next model. The final prediction is a weighted sum of all tree predictions. XGBoost is a scalable and highly accurate implementation of gradient boosting, where trees are built in parallel instead of sequentially. XGBoost follows a level-wise strategy, scanning across gradient values and using these partial sums to evaluate the quality of splits at every possible split in the training set.²⁷ **CatBoost** and **LightGBM** are two other open-source framework for GBDT, used for various tasks at different enterprises. **CatBoost** is developed by Yandex researchers and engineers.²⁸ **LightGBM** is developed by Microsoft.²⁹

NeuralNetTorch is the PyTorch neural network model. PyTorch (*Paszke et al.*, 2019) provides deep neural networks built on a tape-based automatic differentiation system. When building neural networks, this reverse-mode auto-differentiation saves time on one epoch by calculating differentiation of parameters at the forward pass. This allows arbitrarily changing the way the neural network behaves.³⁰ **NeuralNetFastAI** is PyTorch neural network model implemented with the fastai (*Howard and Gugger*, 2020) backend, a deep learning library that simplifies training fast and accurate neural networks.³¹

²⁷Please refer to <https://www.nvidia.com/en-us/glossary/data-science/xgboost/> for details.

²⁸Please refer to <https://catboost.ai> for details.

²⁹Please refer to <https://lightgbm.readthedocs.io/en/latest/> for details.

³⁰Please refer to <https://github.com/pytorch/pytorchmore-about-pytorch> for details.

³¹Please refer to <https://docs.fast.ai> for details.

Appendix E: Identifying bias source at complete UCI adult dataset

We construct the complete 15×15 distance matrix without *sex* excluded from the attribute set. *Sex* has the largest (raw) distance to origin among the 14 attributes (Figure A1a).

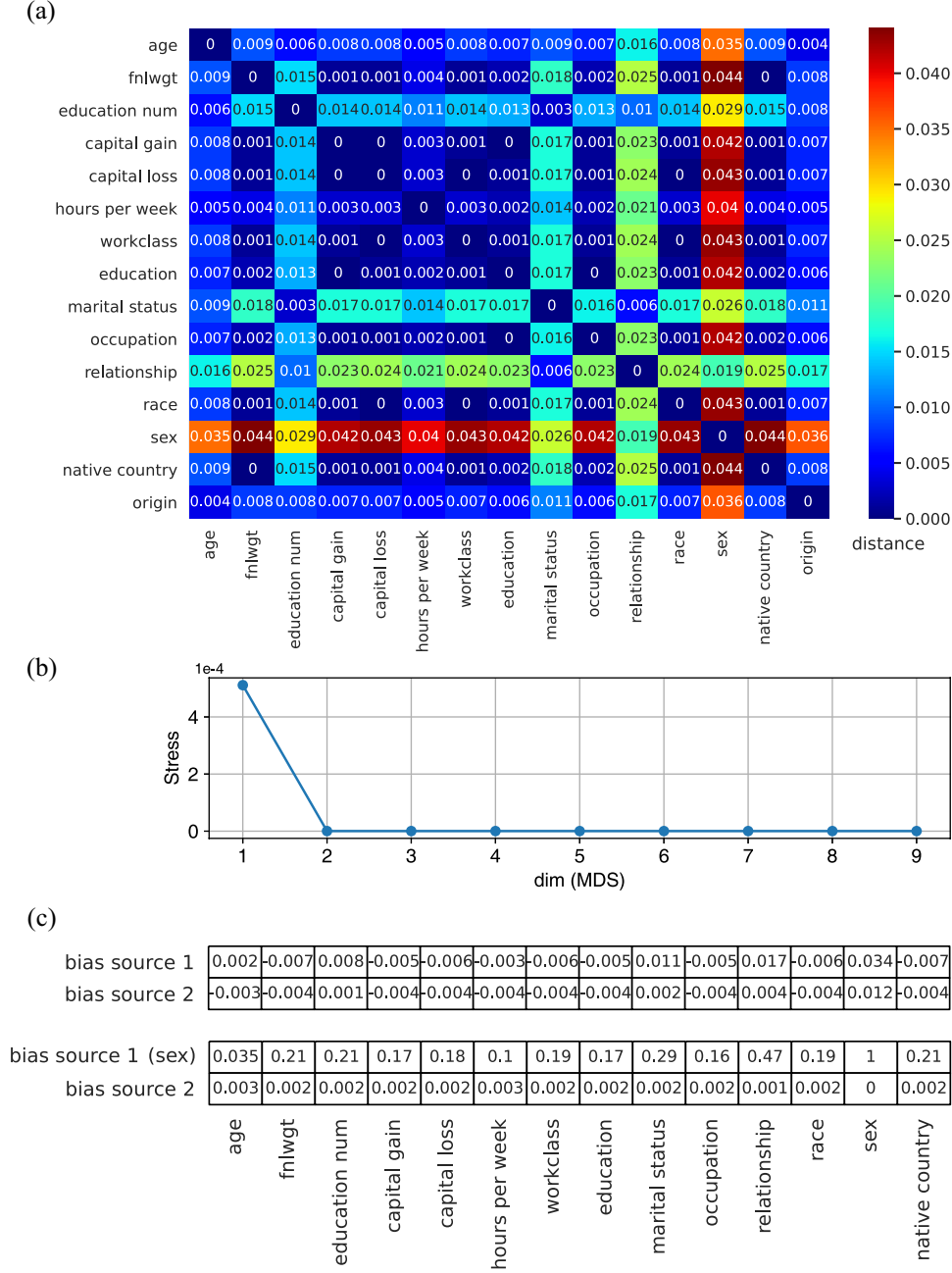


Figure A1: Bias source identification at the complete UCI adult dataset. (a) 15×15 distance matrix. (b) MDS elbow plot. (c) Un-normalized (top) vs. rotation-normalized (bottom) bias concentration matrix.

MDS determines $S = 2$ bias sources (Figure A1b). Projected attribute plane (Figure 8a) confirms *sex*'s high *absolute* bias concentration (without O specified), consistent with prior findings. Rotation-normalized bias concentration matrix (Figure A1c) reveals other attributes' bias concentration at the *sex* (i.e., gender) dimension, consistent with the main result where O is specified as gender.

Appendix F: Protecting different racial groups at UCI adult dataset

With *race* set as *O*, the distance matrix surfaces *sex*, *fnlwgt* and *education num* as having large distance to the remaining cluster of attributes and origin (Figure A2a). MDS determines that $d = 3$ is the appropriate dimension for the attribute space (Figure A2b), and the above three attributes possess the largest bias concentration (distance to origin) in the projected 3-D space (Figure A2c, d).

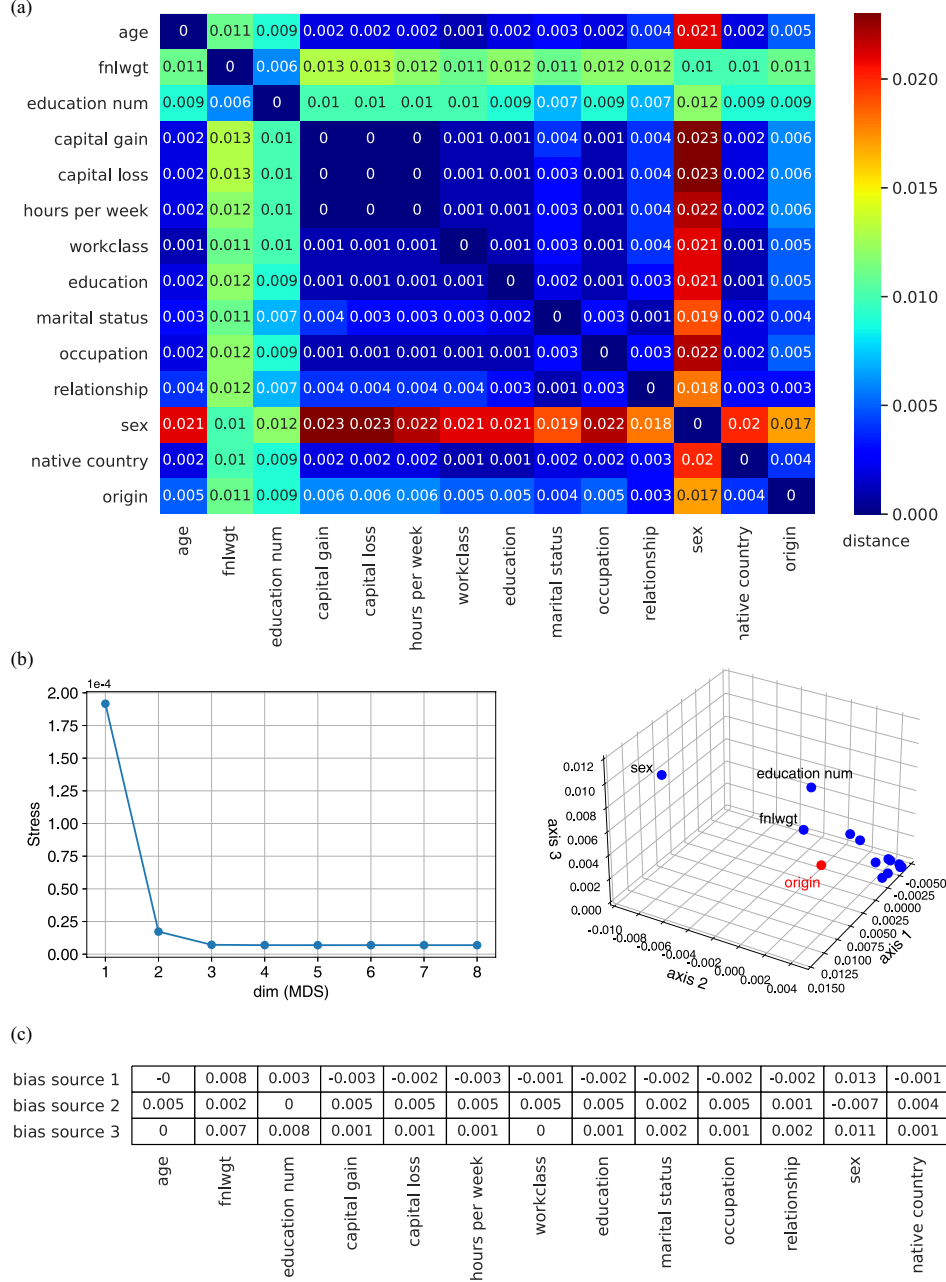


Figure A2: Protecting different racial groups at the UCI adult dataset. (a) Distance matrix. (b) MDS elbow plot. (c) Projected 3-D attribute space. (d) Bias concentration matrix.

Three transformations are needed to confine attributes' bias concentration within $\epsilon = 0.01$ (Table A5): re-binning at *sex* (equivalent to removing it), polynomial transformation ($\alpha = 3$) at *fnlwgt*, and polynomial

transformation ($\alpha = 1/7$) at *education num* (Figure A3a).

$\epsilon = 0.01$			
Iteration	Treated Attribute	Transformation	No. of Trials
#1	sex	re-bin: Male, Female	1
#2	fnlwgt	polynomial: x^3	1
#3	education num	polynomial: $x^{1/7}$	3

Table A5: Bias mitigation iteration at $\epsilon = 0.01$ (protecting different racial groups).

Equality of odds asks that equality of conditional probabilities $\Pr\{\hat{Y} = 1 \mid O = o_i, Y = y\} = \Pr\{\hat{Y} = 1 \mid O = o_j, Y = y\}$ is reached for any two groups o_i, o_j (along the protected attribute O) at a particular label $Y = y$, where Y and $\hat{Y}$ are the actual and the predicted label, respectively. In our setting, similar to statistical parity, fairness of (multi-nary) classification outcome based on equality of odds is noting the maximum pairwise difference in the conditional probabilities, considering multiple categories at O :

$$\Delta_{EO} = \text{Max}_{\text{All } i, j \text{ pairs}} \left\{ \frac{1}{2} \left| \Pr(\text{Income} \geq 50K \mid \text{Race} = i, \text{Income} \geq 50K) - \Pr(\text{Income} \geq 50K \mid \text{Race} = j, \text{Income} \geq 50K) \right| + \frac{1}{2} \left| \Pr(\text{Income} \geq 50K \mid \text{Race} = i, \text{Income} < 50K) - \Pr(\text{Income} \geq 50K \mid \text{Race} = j, \text{Income} < 50K) \right| \right\}. \quad (19)$$

$\Delta_{EO} = 0$ corresponds to perfect equality of odds.

When using Δ_{EO} to indicate classification's fairness, the improvement of fairness is less concerted across the 12 models (Figure A3b) than using statistical parity (Figure 12).

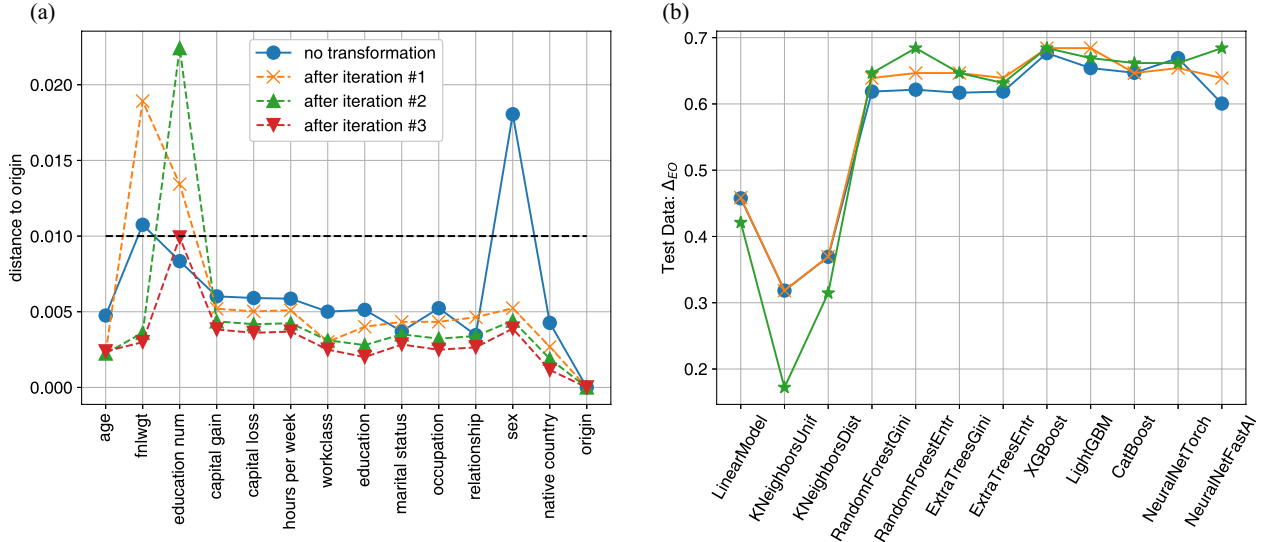


Figure A3: Protecting different racing groups at the UCI adult dataset (continued). (a) Iteration of bias mitigation at $\epsilon = 0.01$. (b) Improvement of classification fairness measured by Δ_{EO} .